((PHAM LANGUAGE .

Released version 202602 )

(Pham language be highest level universal human-AI-AI language .

New feature in version 202602 :

++ Pham language 202602 firstly introduce Pham generalized Boolean axiom and the fundamental Pham undeterministic selection theorem of Pham language fully cover inherent undeterministic content .

++ The innovative Pham-equality-knowledge command is paradigm shift to explore knowledge in AI era .

++ Pham language grammar function repository , Pham OO-translation algorithm , Pham Neighbor-Grouping Translation Algorithm , Pham Anthropic-translation algorithm are innovative AI paradigm shift in 2026y , guarantee lead to AI really think instead of simply list word in probabilistic order as it do up to 202602+ .

Pham language 202602 also introduce absolute rigorous space normalization rule for Pham language text .)

(LICENSE :

Pham language and its architecture and feature as 1 whole human intellectual effort , be designed for optimal use energy consumption of AI , and protect earth environment .

By that reason , it be open-source and free-charged for any use .

But by regard humanity intellectual effort principle , it require at least always call correct term : Pham language , Pham undeterministic selection theorem , Pham generalized Boolean axiom .

Restrict relabel this result by other proper name . Prohibit reupload and republish this result in other form with other name . Because such act violate humanity intellectual effort regard principle .

Any reference or republish must call/refer these correct above term .)

(1 . WHAT BE PHAM LANGUAGE ?

Pham language be the UNIVERSAL HUMAN-AI-AI LANGUAGE . It be the highest level universal human-AI language among all possible existing high level human-AI language .

Human Experience :

Pham language allow all human effectively talk with AI system in human most familiar way .

AI system can effectively understand people .

People can effectively describe precise matter .

AI system use minimal energy for computing .

Human use minimal cognitive load remember Pham language syntax .

In short description :

Human user only require remember very few syntax and command .

Still , human can effectively describe precise matter .

This allow maximal mutual human-AI understanding and require minimal AI energy consumption .

How Use :

Human user simply need use their familiar parent natural language vocabulary ( for example , standard English vocabulary ) with standard mathematical grammar write Pham language code .

As a high level universal AI-AI language , Pham language provide tool and command for AI user share and exchange knowledge .

This tool be independent of any specific AI model .

Example of Pham Language Command :

Example 1 : The title of this text be an example of Pham language command .

Example 2 : ( (look (all file) in (directory 'directoryName')) rename (file with condition ((file name) contain 'abcd')) )

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(2 . PHAM LANGUAGE SPECIFICATION

(2.1 . Core Language Building Strategy

Pham language be an abstract language of human high level universal interface communicate with AI system .

Pham language base on two core component : the worldwide accepted standard English vocabulary , and the worldwide accepted general math grammar .

(2.1.1 . Core Notation Style of Pham Language :

Pham language = (standard English vocabulary + standard mathematical grammar) .

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(2.1.2 . Core Philosophy of Pham Language :

Use standard mathematical model in strict hierarchy structure better specify essence and relationship between essence of any Pham language intention text (code) to the AI system .

Maximally exploit the coincidence between the human user natural language vocabulary and the predefined natural language vocabulary of AI system .

In this specification , we use standard English vocabulary .

Also , maximally exploit the coincidence between worldwide accepted general user knowledge and predefined AI system knowledge in other sector too .

Human user write Pham language code by use mostly their familiar standard English vocabulary .

Human user use well-closed parenthesis group familiar English vocabulary word in different word group .

This follow strict mathematical hierarchy structure .

This specify the intended relationship between word and group of word , and specify the relationship between intended matter .

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(2.1.3 . Mechanism of Understanding :

Because of the coincidence between human brain vocabulary and AI system knowledge , and because of the strict hierarchy structure in well-closed parenthesis , the AI system can resolve Pham language code into suitable interpretation .

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(2.2 . SYNTAX SPECIFICATION

(2.2.1 . The Universal Syntax of Pham Language Code :

Everything in Pham language be in one universal syntax for any Pham language element :

(element1 (element2 (deepNestedElement otherNestedElement)) atomicElement element4 .. elementN)

Everything in Pham language be element .

Element can have nested element .

Element can be inside other containing element .

Every code be many element .

Pham language element be separate by exclusive separator : TangibleSpace (space (UTF-8 0020) , Unix EndOfLine (UTF-8 000A) , Windows EndOfLine (UTF-8 0D0A) , Tab (UTF-8 0009) , none-breaking-space (UTF-8 00A0)) , and parentheses (UTF-8 0028 , UTF-8 0029) .

Use well-closed parenthesis group element in strict hierarchy structure .

TangibleSpace (space (UTF-8 0020) , Unix EndOfLine (UTF-8 000A) , Windows EndOfLine (UTF-8 0D0A) , Tab (UTF-8 0009) , none-breaking-space (UTF-8 00A0)) be not pham language element .

TangibleSpace be element separator in Pham language .

Parentheses (UTF-8 0028 , UTF-8 0029) are not Pham language element .

Parentheses (UTF-8 0028 , UTF-8 0029) be exclusively element separator and grouping symbol in Pham language .

Any appearance of standard parentheses (UTF-8 0028 , UTF-8 0029) in pham language text are automatically reckoned as exclusively grouping symbol and element separator .

Any appearance of standard parentheses (UTF-8 0028 , UTF-8 0029) in pham language text directly impact hierarchy structure of pham language text .

Thus impossible use standard parentheses (UTF-8 0028 , UTF-8 0029) as none-functional parenthesis semantic meaning character in Pham language text .

If the text want hint none-functional parenthesis semantic meaning character , then it must use other proper term to hint none-functional parenthesis semantic meaning character , for example : word 'opening-parenthesis' , 'parenthesis' , 'closing-parenthesis' , or other parenthesis-liked character (for example : CJK parenthesis (UTF-8 FF08 , UTF-8 FF09)) .

All syntax of Pham language be in form of the above 1 universal syntax of any Pham language element .

Pham language dedicatedly have no any other syntax , except the only 1 universal syntax of any Pham language element .

Hierarchy Example :

Atomic Element : the_element

This code be 1 Pham language element . It have no nested element .

Compound Element : (my_element)

This code be 1 element . This element contain 1 nested element : my_element .

Compound Element : ((house of my_name) (black door) (transparent window))

This element have 3 nested element : (house of my_name) , (black door) , and (transparent window) .

This be level-1 of the hierarchy .

Precision and Ambiguity :

User can translate natural language into Pham language with different level of detail .

More grouping create more precision for AI system .

Low Detail : (I want build house which look like my friend house)

High Detail : (I want build (house which (look like) ((my friend) house)))

The high detail version remove ambiguity .

Non-Equivalence of Nesting :

Nesting level change the essence of the matter .

This three code be not equivalent :

element1 Atomic element

(element1) Element contain a nested element .

((element1)) Element contain an element contain a nested element .

Separator define the boundary of matter :

Note : Pham language do not confirm comma , punctuation , quote as pham language element separator . They be usual character in Pham language .

Pham language is simply silent about whether they play as separator in context .

Pham language do not restrict AI system use them as separator in suitable context .

For example : (element1 , element2)

In this example , the comma be a pham language element . There be 3 nested pham language element in this example .

For example : 'A,'

In this example , the comma be usual character and part of 1 element 'A,' .

Because 'A,' be treat as 1 whole element 'A,' .

Comma be not element separator in Pham language .

Logic : In (element1 element2 element3) , the element only see three nested element .

It have no information about a possible combined group (element1 element2) unless you use parenthesis group them .

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(2.3 . ABOUT PHAM LANGUAGE ELEMENT

Everything in Pham language be Pham language element .

Element can be atomic element or compound element .

Atomic Element :

It be not parenthesized .

It have no nested element .

It is sequence of consecutive TangibleSign .

Example : the_word be atomic .

Note : (the_word) be NOT atomic because it have parenthesis .

Compound Element :

It be a well-closed parenthesized element .

Example : (abc) be compound element .

Example : (x y (ab (cd jjj) word) 12) be compound .

Special Case : () be a special compound element .

Convention :

Pham language perceive only qualified element .

Any text , which be not inside a well-closed parenthesis and not be a clean atomic element , be outside the perception of Pham language .

Example : aa =! bb (without parenthesis) be not perceive by Pham language .

Correct : (aa =! bb) be perceive as 1 compound element with 3 nested element .

Each pham language element have its own hierarchy structure .

Example : Code : (lets aa =! bb . Print aa)

In this code example , the AI system implement print aa , thanks to semantic meaning .

But Pham language absolutely not perceive and not know 'aa =! bb' , 'lets aa =! bb' , 'Print aa' , and any analogous stuff , .. in this code .

Pham language perceive only and exclusively only all qualified pham language element in this code : they are : '(lets aa =! bb . Print aa)' , 'lets' , first 'aa' , '=!' , 'bb' , '.' , 'Print' , second 'aa' .

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(2.4 . ABSTRACT RELATIONSHIP BETWEEN ELEMENT

Given Pham language element aa as (aa{1} aa{2} .. aa{n}) .

The hierarchy automatically specify an abstract relationship between nested element aa{i} and aa{j} .

The AI system use its knowledge resolve this abstract relationship into a concrete context-aware relationship .

Example of Context Resolution :

( (my house be here) ( ((friend house) be there) (there be car) (it be large enough park car) ) )

The word it in the last group have an abstract relationship with the nearest level-1 neighbor .

The AI did infer that it dominantly hint friend house because they share the same parent group .

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(2.5 . PHAM LANGUAGE CODEPAGE AND PHAM LANGUAGE MESSAGE

Pham Language Codepage :

This be the content of a text file .

A correct codepage must always be 1 Pham language element (Atomic or Compound) .

The Wrap Convention :

If a file have many element , it must be wrap inside 1 outermost pair of parenthesis .

This ensure the codepage be a single qualified element .

Pham Language Message :

This be a 1 qualified element when it move between application or conversation .

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(2.6 . PHAM INDEX OF ELEMENT

(2.6.1 . Conventional Index :

Given Pham language element aa as (aa_1 .. aa_n) .

Every nested element at level-1 stay in a strict place order .

Pham language 202602 switch use 1-based index for conventional index .

The conventional index of aa_i in aa be simply the positive integer i .

Example : aa be (item bigItem biggerItem (otherItem anotherItem) item)

The first item have index 1 .

bigItem have index 2 .

(otherItem anotherItem) have index 4 .

otherItem have no index in aa because it be not in level-1 .

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(2.6.2 . Pham Index of element inside 1 its Containing Element :

Pham index be an ordered sequence of conventional index that definitely locate one nested element .

It have the form (integer_1 integer_2 .. integer_k) .

Example : aa be (abc 123 (pqr 23a (x y abc z) m) p)

abc (first) be (1)

pqr be (3 1)

x be (3 3 1)

abc (second) be (3 3 3)

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(2.6.3 . Pham Index of element inside 1 its Pham Language Codepage :

In one Pham language codepage , if say Pham index without further specification , then always identify it as the Pham index of element inside the current pham language codepage .

Every element in one codepage have one absolutely unique Pham index .

The codepage itself have no explicit Pham index .

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(2.7 . LOGICAL EXECUTING ORDER OF ELEMENT

In a Pham language codepage , the codepage itself have the smallest (earliest) logical executing order .

Pham language use the Pham index of element for compare the logical executing order of element .

Logical executing order of element be not always measure define real physical executing (interpreting) order of element in Pham language codepage .

(2.7.1 . Pham language comparison principle for logical executing order of element :

Pham language comparison principle for logical executing order define rule to compare 2 logical executing order of element

Given element A with Pham index I = (i_1 .. i_n) and element B with Pham index J = (j_1 .. j_m) .

Case 1 : The Shared Path (Parent and Child) :

If (i_1 .. i_n) be the same as the start of (j_1 .. j_m) and m > n , then element A be lower order than element B .

Example : I = (1 2) , J = (1 2 3) . Here , I be lower order .

Case 2 : The First Difference (Left Right) :

Find the first position p where i_p and j_p be different .

If i_p < j_p , then I be lower order than J .

If i_p > j_p , then I be higher order than J .

Example : I = (1 5 9) , J = (1 7 2) . Because 5 < 7 , I be lower order .

Example : I = (8 2) , J = (7 9 3) . Because 8 > 7 , I be higher order .

Case 3 : Matter Identity :

If n = m and all integer be equal , then I and J be the same element .

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(2.7.2 . Property of logical executing order of element in pham language codepage :

In a pham language codepage , Pham index of element be the measure of logical executing order of element .

In a pham language codepage , because each element have its unique Pham index , thus each element have its unique logical executing order , which never coincide with logical executing order of other element .

In a pham language codepage , because logical executing order of each element be unique and ordered , thus can arrange logical executing order of element in an ordered numbered sequence with integer ordering number .

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(2.8 . PLACE FLAT ORDER OF ELEMENT :

In a pham language codepage , the pham language codepage itself have smallest logical executing order .

Thus by convention , reckon place flat order of the pham language codepage 0 .

After the pham language codepage , the next element with smallest logical executing order will have place flat order as 1 .

After the element with place flat order as 1 , the next element with smallest logical executing order will have place flat order as 2 . And so on .. .

Thus , in a pham language codepage , use logical executing order of element as ordering criterion , can arrange all element of the pham language codepage in 1 place flat ordered sequence with numbering 0 , 1 , 2 , .. .

The pham language codepage itself stay at position with order number 0 .

Place flat order of element be not always measure define real physical executing (interpreting) order of element in Pham language codepage .

For example :

The pham language codepage : (ab cd (a word (another word) something) pqrs)

In this example , can arrange all element as follow :

(ab cd (a word (another word) something) pqrs) : place order : 0 . Logical executing order : smallest .

ab : place order : 1 . Logical executing order : (1) .

cd : place order : 2 . Logical executing order : (2) .

(a word (another word) something) : place order : 3 . Logical executing order : (3) .

a : place order : 4 . Logical executing order : (3 1) .

word : place order : 5 . Logical executing order : (3 2) .

(another word) : place order : 6 . Logical executing order : (3 3) .

another : place order : 7 . Logical executing order : (3 3 1) .

word : place order : 8 . Logical executing order : (3 3 2) .

something : place order : 9 . Logical executing order : (3 4) .

pqrs : place order : 10 . Logical executing order : (4) .

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(2.9 . PHAM LANGUAGE TEXT AND ENCODING

(Annex2 of this specification contain Python code for all qualified pham language char) .

Pham language text be UTF-8 encoded text .

Pham language text is ordered string of exclusively Pham language TangibleChar .

Pham language TangibleChar is : TangibleSign or TangibleSpace or parenthesis , and nothing else .

Pham language text filtering rule :

Pham language text contain only Pham language TangibleChar

Pham language text ignore all character , except Pham language TangibleChar

Pham language text filtering normalization procedure :

Replace any character , which be not TangibleChar , by empty '' .

Definition of TangibleSpace :

TangibleSpace be character that create non-zero-width whitespace in text file .

It only be one of the 5 following character :

(space (UTF-8 0020) , Unix EndOfLine (UTF-8 000A) , Windows EndOfLine (UTF-8 0D0A) , Tab (UTF-8 0009) , none-breaking-space (UTF-8 00A0))

Definition of TangibleSign :

TangibleSign be character that strictly satisfy all five condition below :

Condition 1 : It be exactly one single code point character .

Condition 2 : It result in exactly one 16-bit code unit in UTF-16 encoding (it stay in Basic Multilingual Plane) .

Condition 3 : It belong strict Unicode category : Letter , Number , Punctuation , Symbol , or visible Mark .

Condition 4 : It must print non-empty-pixel image . It must not be invisible mark , zero-width joiner , or zero-width space .

Condition 5 : It be not parenthesis .

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(2.10 . SPACE NORMALIZATION PROCEDURE FOR PHAM LANGUAGE TEXT :

To ensure the AI system perceive every Pham language element without ambiguity and with optimal energy efficiency , normalized Pham language text must follow strict space normalization procedure .

Step 1 : Space Conversion

If arbitrary character be TangibleSpace , replace it by one single standard space .

Step 2 : Space Sequence Collapse

Replace any sequence of consecutive space by exactly one single space .

Step 3 : Parenthesis and Sign Spacing

Apply the following replacement rule define the exact boundary of matter :

Rule 1 : Remove space between opening parenthesis and opening parenthesis :

Illustration : "( (" become "((" .

Rule 2 : Remove space between opening parenthesis and TangibleSign :

Illustration : "( tangibleSign" become "(tangibleSign" .

Rule 3 : Add one single space between TangibleSign and opening parenthesis :

Illustration : "tangibleSign(" become "tangibleSign (" .

Rule 4 : Remove space between closing parenthesis and closing parenthesis :

Illustration : ") )" become "))" .

Rule 5 : Remove space between TangibleSign and closing parenthesis :

Illustration : "tangibleSign )" become "tangibleSign)" .

Rule 6 : Add one single space between closing parenthesis and TangibleSign .

Illustration : ")tangibleSign" become ") tangibleSign"

Rule 7 : Add one single space between closing parenthesis and opening parenthesis :

Illustration : ")(" become ") ("

(Note : This specification text itself is pham language text .

Thus this specification text use other-parentheses (UTF-8 2768 , UTF-8 2769) in the above illustration to illustrate standard parenthesis in space normalization rule .

Never put unfunctionalize standard parentheses (UTF-8 0028 , UTF-8 0029) in Pham language text .)

For example : ( ( ( distribute ( economy profit ) to ( every human ) ) ) because ( ( human money ) be ( ( root client finance source ) of economy ) ) )

In this example , space normalization procedure give result : (((distribute (economy profit) to (every human))) because ((human money) be ((root client finance source) of economy)))

Pham language space normalization procedure not require that pham language text must be in well-closed parenthesis .

Pham language space normalization procedure apply on any pham language text .

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(2.11 . DEFINITE NUMBER OF ELEMENT PRINCIPLE

(2.11.1 . Definite number of element of pham language codepage :

Every concrete Pham language codepage , Pham language command , or Pham language message must contain exactly one definite number of element .

One arbitrary concrete Pham language text never contain infinite number of element .

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(2.11.2 . Semantic Capability of Codepage :

Although one Pham language codepage must contain one definite number of element , it can describe one mathematical set of infinite number of element .

Example of Semantic Capability :

Code : (set of (all (positive integer)))

Physical Structure : This code contain exactly 7 nested Pham language element . It have one very small , definite number of element .

Semantic Meaning : This code perfectly describe one mathematical concept that contain infinite number of integer (1 , 2 , 3 , 4 , ..) .

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(2.12 . STANDARD ENGLISH VOCABULARY PRIORITY PRINCIPLE

(2.12.1 . Priority of Standard English Vocabulary :

(This specification be for integrated English version of Pham language . Specification for integrated other natural language version of Pham language be analogous .)

Human user must give priority use standard English vocabulary ( standard English word ) .

When human user use standard English vocabulary , it guarantee that all AI system can exploit predefined worldwide accepted meaning of standard English word .

This allow AI system exactly generate intended interpretation of Pham language command .

Typical Pham language command not specify concrete meaning of word .

It only specify two thing :

Existence of element in explicit mathematical hierarchy structure .

Abstract relationship between element in the same level of hierarchy structure .

The AI system read English word in Pham language command , and see explicit hierarchy structure know element abstract relationship .

Then AI system use rich arsenal of predefined worldwide accepted meaning of standard English vocabulary , in combination with strict hierarchy structure , generate final interpretation .

Final interpretation can be text answer , image , video , or explicit programming code ( like Python code , Rust code , C code ) .

Thus , priority use standard English vocabulary guarantee that AI system can quickly understand Pham language command .

Human user can give definition weird English keyword , but it be not good practice , because AI system knowledge about weird English keyword be poor .

Every word that have intended meaning for AI system and other people understand should be standard English word .

To specify a unique private name of an element , human user can use private word outside standard English vocabulary .

Example :

The word 'myPrivateHouse' be not in standard English vocabulary .

It be use identify name of private thing .

( my house ( name = 'myPrivateHouse' , type be townhouse , color = green ) )

( generate image ( ( 2 myPrivateHouse in row ) in sunshine ) )

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(2.12.2 . Word can also keep its infinitive form in plural form :

Because adjacent numeric words can specify plurality , thus main word can keep its infinitive form in plural form .

For example :  ((2 car) =! (2 cars)) in many context in Pham language .

For example : ((3 child) =! (3 children)) in many context in Pham language .

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(2.12.3 . Levelize Human Cognitive Load with Short Text Record :

More explicitness of Pham language command equal less required computing energy in AI system .

Shorter keyword create shorter text record , but it require extreme big additional human cognitive load when human user or other people re-read Pham language message next time .

Human user should levelize this relation .

Human user should avoid too short weird keyword in Pham language command .

Short weird keyword can save text typing , but it mostly not save machine code realization time , and it destroy the core benefit of minimal human cognitive load .

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(2.12.4 . Universal AI Inference Principle :

In Pham language , many recommended syntax form be not rigid specification . AI system be allow automatically infer other syntax form of Pham language command .

This principle not reduce universality of Pham language . Explanation :

All capable AI system share the same worldwide accepted general knowledge base ( comprising worldwide accepted standard English vocabulary meaning , worldwide accepted school and university subject knowledge , worldwide accepted programming language knowledge , and worldwide accepted scientific knowledge ) .

Because this worldwide accepted general knowledge base be identical across all capable AI system , inference by any capable AI system on any standard Pham language command with standard English vocabulary always produce the same interpretation .

Thus , non-rigid recommended syntax form in Pham language , when use with standard English vocabulary , be universally deterministic across all capable AI system .

Only inadequate AI system , which not load worldwide accepted general knowledge , can produce deviant inference . Such deviant AI system be not qualified for Pham language use . Pham language specifically target capable AI system .

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(2.13 . GENERAL RULE FOR PHAM LANGUAGE COMMAND

One arbitrary Pham language command be always one whole Pham language element .

Thus , almost every Pham language command must be wrap inside one outermost well-closed parenthesis pair , because one Pham language compound element be always inside one outermost well-closed parenthesis pair .

One Pham language command must always be one well-closed parenthesis Pham language element , except when the command be one atomic element .

Pham language perceive only pham language element .

Pham language text can content many (pham language text part) , which be not pham language element .

Pham language not perceive any (pham language text part) , which be not pham language element .

RIGID REMINDER : IF THE PHAM LANGUAGE COMMAND NOT ATOMIC ELEMENT , THEN THE PHAM LANGUAGE COMMAND MUST ALWAYS AND ALWAYS STAY INSIDE ITS OWN OUTERMOST WELL-CLOSED PARENTHESIS PAIR . OTHERWISE , PHAM LANGUAGE SIMPLY NOT PERCEIVE THIS COMMAND .

Example 1 :

(aa put bb on table)

In this example , '(aa put bb on table)' be 1 pham language command , because it be 1 pham language element .

But 'aa put bb' be not 1 pham language command , Pham language even not perceive the text 'aa put bb' inside '(aa put bb on table)' .

Pham language know only pham language element : 'aa' , 'put' , 'bb' , 'on' , 'table' , and '(aa put bb on table)' .

Example 2 :

((aa put bb) on table)

In this example , '(aa put bb)' be 1 pham language command , because it be 1 pham language element .

(2.13.1 . Pham executing order , logical executing order of (command definition) :

Each arbitrary pham language element can be 1 possible Pham language command .

Each Pham language command must always be strictly 1 whole Pham language element .

In a Pham language codepage , Pham language command aa can play as definition of command_or_rule bb .

In a Pham language codepage , each user defined command_or_rule bb must be always generated from some command-definition command aa in the Pham language codepage .

Definition : Pham executing order of command-definition :

Pham executing order of command-definition of command_or_rule bb is the Pham executing order of aa , where aa is the command-definition for command_or_rule bb .

Definition : Logical executing order of command-definition :

Logical executing order of command-definition of command_or_rule bb is the logical executing order of aa , where aa is the command-definition for command_or_rule bb .

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(2.13.2 . Pham language universal command-conflict resolution principle :

When AI system interpret concrete element examined_element , the AI system sometimes can meet command-conflict-case :

((examined_element fall in case of

((must apply command_or_rule aa) and

(must apply command_or_rule bb) and

(command_or_rule aa conflict with command_or_rule bb))))

Pham language universal command-conflicted resolution principle specify resolution for such conflicted command-conflict case :

((Priority1 : AI system use its data of worldwide accepted general knowledge and aware context to infer decide whether to choose aa or bb to apply .)

(Priority2 : Choose the command_or_rule with higher Pham executing order of its command-definition to apply . But if (mutual conflicted command_or_rule aa and command_or_rule bb) have same Pham executing order of command-definition , then ((do not apply aa for the examined_element) and (do not apply bb for the examined_element)))

(Priority3 (lower priority): If AI system can not define Pham executing order of command-definition of these conflicted command , and of course these conflicted command are not identical , then apply priority :

canonical assingment (description in further part) have higher applying priority than (command_or_rule of other type)

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(2.14 . ASSIGNMENT COMMAND IN PHAM LANGUAGE : (:=)

(Specified term : canonical assignment syntax)

Pham language recommend use popular canonical word := for assignment operation :

((interested stuff) := (target stuff))

(2.14.1 . Innovation of Compound Label-Reference :

The special assignment concept , firstly introduce in Pham language 202602 2026y , be drastically more flexible than assignment concept in all popular programming language up to year 2026 .

In all popular programming language up to year 2026 , label-reference of assignment must be one atomic element .

But in Pham language , the label-reference (interested stuff) can be one arbitrary Pham language element ( atomic or compound ) .

The requirement of allowance use one compound Pham language element as label-reference be one rigid unique Pham language specific specification .

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(2.14.2 . Syntax Structure of Assignment :

In this recommended syntax , note that the whole assignment command be also inside exactly one pair of outermost well-closed parenthesis .

The whole assignment command be exactly one whole Pham language compound element , which contain strictly 3 nested element : (interested stuff) , ':=' , and (target stuff) .

Example 1 : Code :

((the form of effort) := ((labor) then (test) then (again labor) then (again test) then (so on) ..))

Example 2 : Code :

(suppose that aa := bb , do something)

In this example , there is not Pham language specific assignment specification . Even though , AI system can infer assignment intention , but it is AI specific assignment operation .

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(2.14.3 Canonical syntax emphasis :

Form (aa := bb) is the canonical assignment command syntax in Pham language .

There be other analogous synonym syntax form for assignment operation . When AI system meet such analogous form , AI system use its data of worldwide accepted general knowledge and context awareness to infer and decide .

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(2.14.4 Pham language assignment scope specification :

The AI system automatically infer scope of (assignment command) by (AI knowledge and aware context) .

Definition : The current working pham language codepage be the pham language codepage that physically contain the assignment command .

Pham language specific instruction : (if (the AI system can not find any context-aware argument for infer scope of (assignment command)) then (the AI system reckon scope of (assignment command) as the current working pham language codepage)) .

(Example :

Variable Assignment Code :

((my beautiful House) := ((Name : Omega) (type : good House) (have garden) stay along the road))

Question :

(What be (my beautiful House) ?)

In this example , result : ((Name : Omega) (type : good House) (have garden) stay along the road) .)

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(2.14.5 Pham language assignment application priority specification :

Suppose that AI system already implemented (the concrete assignment rule) of pham language canonical syntax assignment command .

AI system use its data of worldwide general accepted knowledge , and aware context , to infer the priority of applying for (the concrete assignment rule) .

If AI system do not have any clue to infer priority of applying for (the concrete assignment) , then AI system can use Pham language specification : take (pham language canonical assignment rule) with higher Pham executing order of its command-definition to apply . But if these conflicted Pham language canonical assignment rule have equal Pham executing order of command definition , then the AI system do not apply any of them .

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(2.14.6 Resolve conflict of Pham language assignment command :

if (assignment (aa := bb) with its established scope) conflict with (assignment (aa := cc) with its established scope) , then the AI system use its data of worldwide accepted general knowledge and aware context to infer resolve .

(if the AI system can not find any clue to resolve conflict , then the AI system follow Pham language specification :

should prioritize assignment , which be declared later (order by Pham executing order of assignment command) .

If these 2 conflicted assignment-insertion-command have same Pham executing order , and of course these 2 conflicted assignment-insertion-command are not identical , then AI system do not apply any of them in this case .)

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(2.15 . PHAM-EQUALITY-KNOWLEDGE COMMAND (=!) :

(Specified term : Pham-equality-knowledge)

Pham language syntax for Pham-equality-knowledge command :

((stuff aa) =! (stuff bb))

Pham-equality-knowledge concept be innovative and fundamental . It be the primary bridge between human-defined truth and AI-driven reasoning .

Unlike standard assignment (:=) , which mandate immediate and universal replacement of a label by its value , Pham-equality-knowledge (=!) establish a Non-Coercive Identity .

When a user declare (aa =! bb) , they announce a new state of knowledge to the AI system .

The AI system accept this identity to respect user intent , but it retain autonomy to choose if and when to apply this knowledge base on the specific requirement of the task .

In the command (aa =! bb) , aa and bb can be pham language atomic element or compound element .

(2.15.1 . The Principle of Non-Forced Substitution :

The profound power of Pham-equality-knowledge stay in its flexibility .

Fact1 . Contextual Application : The AI system can substitute aa by bb in scenario where the substitution simplify the problem or provide necessary detail .

Fact2 . Identity Preservation : The AI system can choose not to touch the knowledge , allow aa to remain in its original form to preserve the native text clarity or to avoid unnecessary computational complexity .

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(2.15.2 . Case Study : Newton Second Law :

The superiority of Pham-equality-knowledge be demonstrate by physical formula .

(User Knowledge Input : (Force =! (mass * acceleration)) )

Application A : In complex trajectory calculation , the AI system can flexibly replace 'Force' by (mass * acceleration) to realize detail calculation .

Application B : In standard physical description , the AI system can keep the element 'Force' untouched to preserve human readability .

Comparison : If user use assignment (Force := (mass * acceleration)) , then the AI system be force to replace every 'Force' by (mass * acceleration) , which lead to unnecessary complex and unsolvable expression .

)

(2.15.3 . Bidirectional Application :

Because Pham-equality-knowledge be an acknowledgement of identity rather than a directional data-flow , the AI system treat the relationship as symmetrical :

Fact1 : Forward Substitution : Replace aa by bb ( expand a concept into its detail component ) .

Fact2 : Reverse Substitution : Replace bb by aa ( simplify a complex expression into a concise label ) .

)

(2.15.4 . Distinction from Boolean Equality :

A critical distinction must be make between Pham-equality-knowledge (=!) and standard Boolean-equality-query (==) .

Fact1 : The Boolean Failure : (Force == (mass * acceleration)) result in immediate evaluation and the formula disappear into a single bit of data (TRUE or FALSE) .

Fact2 : The Pham Persistence : Pham-equality-knowledge (=!) ensure the relationship persist as an active structural link in the AI knowledge buffer for future reasoning .

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(2.15.5 . Autonomous Knowledge Integration :

Pham-equality-knowledge (=!) be a True Knowledge Insertion .

Once a user announce (aa =! bb) , the AI system be authorize to use this identity to solve or improve its internal knowledge gap in the background , even if the user do not request to do this .

The AI system can continuously refine its internal world-model and fill in the blank in its background processing use the Pham-equality-knowledge bridge .

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(2.15.6 . Pham-equality-knowledge scope and respect :

The AI system automatically infer scope of (Pham-equality-knowledge command) by (AI knowledge and aware context) .

(Definition : The current working pham language codepage be the pham language codepage that physically contain the assignment command .

Pham language specific instruction : (if (the AI system can not find any context-aware argument for infer scope of (Pham-equality-knowledge command)) then (the AI system reckon scope of (Pham-equality-knowledge command) as the current working pham language codepage)))

(Pham-equality-knowledge acknowledge that the AI exist in a collaborative state with the human user .

The AI receive the knowledge (aa =! bb) as a steady confirmation of the user worldview .

The AI be not require to verify the absolute objective correctness of the knowledge ; rather , it use the knowledge to align its reasoning with the user specific context . This protect the Absolute Logical Abstractness of the system .)

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(2.15.7 . Canonical Syntax Emphasis of Knowledge Insertion :

Form (aa =! bb) be the canonical Pham-equality-knowledge command syntax.

There be other analogous synonym syntax form for equality-knowledge insertion. When AI system meet such analogous form, AI system use its data of worldwide accepted general knowledge and context awareness to infer and decide.)

(2.15.8 . Pham-equality-knowledge Application Priority Specification :

Suppose that AI system already implemented the concrete identity of a Pham-equality-knowledge canonical syntax command .

AI system use its data of worldwide general accepted knowledge, and aware context, to infer the priority of applying this identity for background synthesis or reasoning.

(If AI system do not have any clue to infer priority, then AI system follow Pham language specification: And Pham-equality-knowledge have higher priority in determining semantic identity over standard probabilistic inference.))

(2.15.9 . Resolve Conflict of Pham-equality-knowledge Command :

If a knowledge insertion (aa =! bb) with its established scope conflict with another insertion (aa =! cc) within the same scope, then the AI system use its data of worldwide accepted general knowledge and aware context to resolve.

(If the AI system can not find any clue to resolve conflict, then the AI system follow Pham language specification: Should prioritize the knowledge insertion which be declared later (ordered by Pham executing order). This ensures the AI adapts to the most recent user-provided truth.

If these 2 conflicted knowledge insertion have same Pham executing order , then AI system do not apply any of them .))

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(2.16 . BOOLEAN-EQUALITY-QUERY COMMAND :

(Specific term : boolean-equality-command)

Pham language recommend use canonical syntax for Boolean-equality-query command :

((stuff aa) == (stuff bb))

If (stuff aa) be equal (stuff bb) , then (((stuff aa) == (stuff bb)) =! TRUE) .

In this syntax , note that whole Boolean-equality-query command be also inside exactly 1 pair of outermost well-closed parenthesis .

Whole Boolean-equality-query command be exactly 1 whole Pham language compound element , which contain strictly 3 nested element : (stuff aa) , '==' , and (stuff bb) .

(2.16.1 . Canonical Syntax Emphasis of equality-query command :

Form (aa == bb) be the canonical boolean-equality-query command syntax.

There be other analogous synonym syntax form for equality-knowledge insertion. When AI system meet such analogous form, AI system use its data of worldwide accepted general knowledge and context awareness to infer and decide.)

(2.16.2 Pham language equality-relativity principle :

Equality type is not rigid . The AI system must automatically infer type of equality by use AI general knowledge and aware context .

The AI system must carefully levelize not infer type of equality become so far concrete context .

Example of Boolean-Equality-Query Command :

Example 1 : Mathematical Context Code :

(print((10 / 2) == 5))

In this example , AI system print true . AI system easily infer strict mathematical numeric equality .

Example 2 : State Query Context Code :

( ((current machine state) := idle) (print((current machine state) == running)) )

In this example , whole code be wrap in 1 outermost parenthesis establish strict abstract relationship .

Every element follow strict Space Normalization Procedure . AI system evaluate query and print false .

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(2.17 . Compare Pham-equality-knowledge , boolean-equality-query , assignment)

(2.17.1 . Comparison of Operational Essence :

( (Assignment (aa := bb)) : (A Substitution Command which defines aa as a reference for bb . It mandates that the AI system replace the reference with its assigned value during execution) )

( (Pham-equality-knowledge (aa =! bb)) : (A Knowledge Insertion which announces the semantic identity of aa and bb . It creates a persistent , bidirectional bridge in the AI system internal world-model) )

( (Boolean-equality-query (aa == bb)) : (An Evaluation Query which asks the AI system to compare the current state of aa and bb . It returns a transient Boolean value and do not alter the AI system internal knowledge buffer) )

)

(2.17.2 . Comparison of Persistence and Force :

( (Persistence : ( (Assignment : steady until redefined) (Pham-equality-knowledge : permanent and active in background) (Boolean-query : transient and forget after evaluation) ) ) )

( (Coercion : ( (Assignment : Mandatory substitution) (Pham-equality-knowledge : Flexible contextual substitution) (Boolean-query : No substitution) ) ) )

( (Directionality : ( (Assignment : Unidirectional (aa -> bb)) (Pham-equality-knowledge : Bidirectional (aa <-> bb)) (Boolean-query : None) ) ) )

)

(2.17.3 . Comparison of AI Internal Behavior :

( (Logic of Assignment) : (The AI system prioritize data movement . If (Force := (mass * acceleration)) then 'Force' lose independent identity and become a container for the math) )

( (Logic of Pham-equality-knowledge) : (The AI system prioritize Understanding . If (Force =! (mass * acceleration)) then the AI system understand the shared essence . It can autonomously repair knowledge gaps or simplify complex outputs (Reverse Substitution) without direct command) )

( (Logic of Boolean Query) : (The AI system prioritize Verification . If (Force == (mass * acceleration)) then the AI system only checks if current values match . The relationship be not held for future reasoning) )

)

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(2.17.4 The freestyle-equality command (=) :

Canonical syntax : (aa = bb)

Pham language is silent about (aa = bb) .

The AI system in its freestyle interpret/execute the freestyle-equality command

)

(2.18 . Conditional-Branching Command

Pham language recommend use canonical syntax for conditional-branching command :

Recommended Form 1 :

(if (condition) then (subCommand_1 subCommand_2 .. subCommand_n))

Recommended Form 2 :

(if (condition) then (subCommand_1 .. subCommand_n) else (otherCommand_1 .. otherCommand_m))

In both example , note that the whole conditional-branching command be exactly inside one pair of outermost well-closed parenthesis .

The whole command be exactly one whole Pham language compound element .

(Important : Canonical syntax emphasis :

The above syntax is the canonical conditional-execution command syntax in Pham language .

There be other analogous synonym syntax form for conditional-execution command . When AI system meet such analogous form , AI system use its data of worldwide accepted general knowledge and context awareness to infer and decide .)

Mechanism of Understanding :

Human user use their memory of familiar execution syntax from general knowledge , school subject , or popular programming language .

Human user must use well-closed parenthesis specify exactly one whole Pham language element .

This allow AI system exactly and quickly allocate the specified element for conditional execution .

Because AI system already load worldwide standard knowledge , AI system easily guess and understand human user intention .

Example : Code :

((printTask_x := (print (OK x))) (x := 3) (if (x > 0) then (printTask_x)))

In this example , AI system will print : OK 3 .

Pham language not rigidly require that the syntax (if (x > 0) then (printTask_x)) be conditional execution .

It be user syntax recommendation .

But this syntax pattern appear in natural English language and programming language .

AI system remember this load knowledge and interpret intended conditional execution .

The primary task of human user be not memorize Pham language specific syntax , but strictly group entity form element (some condition) and (do something) in one pham language universal syntax .

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(2.19 . LOOP COMMAND IN PHAM LANGUAGE

Pham language recommend use canonical syntax for loop command :

For-Loop Form 1 :

(for (item in (some set)) do (subCommand_1 .. subCommand_n))

For-Loop Form 2 :

(for (item from p to k) do (subCommand_1 .. subCommand_n))

While-Loop Form :

(while (condition) do (subCommand_1 .. subCommand_n))

Repeat-Loop Form :

(repeat (subCommand_1 .. subCommand_n) until (condition))

In every example , note that the whole loop command be inside exactly one pair of outermost well-closed parenthesis .

The whole loop command be exactly one whole Pham language compound element .

(Important : Canonical syntax emphasis :

The above syntax is the canonical loop command syntax in Pham language .

There be other analogous synonym syntax form for loop command . When AI system meet such analogous form , AI system use its data of worldwide accepted general knowledge and context awareness to infer and decide .)

Example 1 ( Direct Execution ) : Code :

(for (i from 1 to 3) do (print i))

In this example , AI consequently print number 1 , then number 2 , then number 3 .

AI system see this familiar syntax and easily guess execution loop .

The extremely important task of human user be rightly group entity in right Pham language element specify right relationship .

If human user wrongly group entity , no AI system can help .

Example 2 ( Code Generation ) : Code :

( ((this pham language message) be generate (Python code)) (myList =! (a list object)) (for (temporaryFile in (directory 'MyDirectory')) do (add (name of temporaryFile) to myList)) )

In this example , the loop command not make direct execution loop .

It specify AI system create Python code that iterate file and save file name to myList .

The loop command use familiar syntax pattern .

AI system see this syntax , easily guess , and rightly interpret human user intended execution loop for code generation .

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(2.20 . FUNCTION-DECLARATION COMMAND AND FUNCTION-CALLING COMMAND

(2.20.1 . Function-Declaration Command :

Pham language recommend use canonical syntax for function-declaration command :

( define function ( (describe function name) (describe input parameter (declare input_1) .. (declare input_n)) (description of return value) ((implementation_command_1 .. implementation_command_m)) ) )

(Canonical syntax emphasis :

The above syntax is the canonical function-declaration command syntax in Pham language .

There be other analogous synonym syntax form for function-declaration command . When AI system meet such analogous form , AI system use its data of worldwide accepted general knowledge and context awareness to infer and decide .)

The special function-declaration command , firstly introduce in Pham language 202602 2026y , be drastically more flexible than function-declaration command in all popular programming language up to year 2026 .

In all popular programming language up to year 2026 , function name must be one atomic element .

But in Pham language , (function name) can be one Pham language element ( atomic or compound ) .

In Pham language , in the function declaration , function input argument can be also pham language compound element too .

Human user must carefully choose (function name) so that it not conflict with other assignment .

In this syntax , note that the whole function-declaration command be also inside exactly one pair of outermost well-closed parenthesis .

The whole command be exactly one whole Pham language compound element .

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(2.20.2 . Function-Calling Command :

Pham language recommend use canonical syntax for function-calling command :

((function name) (input parameter))

In this syntax , note that the whole function-calling command be also inside exactly one pair of outermost well-closed parenthesis .

The whole function-calling command be exactly one whole Pham language compound element , which contain first nested element as (function name) .

Innovation in 2026y : (function name) can be one atomic element or one compound element .

Energy-Saving Trade-Off :

Other programming language use regular expression for function-calling command .

Pham language firstly in year 2026 use universal syntax for function-calling command allow flexible (function name) and avoid regular expression .

It have trade-off : human user need type additional parenthesis character , but it allow AI system drastically save computing energy .

AI system avoid regular expression and consistently work in only one Pham language universal syntax when it deal with big number of function-calling command .

If there be no previous function declaration with this (function name) , then ((function name) (input parameter)) simply be reckon as usual Pham language element .

Ordering rule : Function-calling command must always appear after its corresponding function-declaration command in codepage order . AI system not recognize any function-calling command that appear before its function-declaration command in codepage order . Pham language not support forward declaration .

(Important : Canonical syntax emphasis :

The above syntax is the canonical function-calling command syntax in Pham language .

There be other analogous synonym syntax form for function-calling command . When AI system meet such analogous form , AI system use its data of worldwide accepted general knowledge and context awareness to infer and decide .)

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(2.20.3 Pham language function-declaration scope specification :

The AI system automatically infer scope of (function-declaration) by (AI knowledge and aware context) .

Definition : The current working pham language codepage be the pham language codepage that physically contain the function-declaration command .

Pham language specific instruction : (if (the AI system can not find any context-aware argument for infer scope of (function-declaration command)) then (the AI system reckon scope of (function-declaration) as the current working pham language codepage)) .

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(2.20.4 Resolve conflict of Pham language function-declaration command :

if (function-declaration command FD1 with its established scope) conflict with (function-declaration command FD2 with its established scope) , then the AI system use its data of worldwide accepted general knowledge and aware context to infer resolve .

(if the AI system can not find any clue to resolve conflict , then the AI system follow Pham language specification :

should prioritize (function-declaration command with its established scope) , which be declared later (order by Pham executing order of (function-declaration command with its established scope)) .

If these 2 (function-declaration command with its established scope) have same Pham executing order , then AI do not do any of them .)

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(2.20.5 . Example of Function Declaration and Calling :

Example 1 : Pure Universal Syntax ( No Regular Expression )

Function declaration with detailed explanation :

( define function ( (function name : (my task)) (input parameter : ( )) (function implementation : (repeat (print Hello) (3 time))) ) )

Same function declaration with less detailed explanation :

(define function ((my task) () (repeat (print Hello) (3 time))))

Function calling :

((my task) ())

In this example , AI system not use regular expression recognize function declaration and function calling .

Note : If there be no previous declaration for (my task) , then AI system treat ((my task) ()) as usual element .

Example 2 : Familiar Math Syntax ( Require Regular Expression )

Code :

((my Function)[x , y] := (x + y))

Function-calling command :

(print (my Function)[1 , 2])

In this example , AI system print number 3 .

Pham language not rigidly specify this syntax , but this syntax appear in school subject or programming language .

AI system already load this standard knowledge , see this syntax , and easily guess human user intention .

In this example , AI system must use regular expression recognize function declaration and function calling .

Example 3 : Familiar Programming Syntax ( Require Regular Expression )

Code :

(define function myFunction{x , y} := (x * y))

Function-calling command :

(print myFunction{2 , 3})

In this example , AI system print number 6 .

AI system use load standard knowledge guess and interpret user function declaration .

AI system must use regular expression .

Example 4 : Grouped Programming Syntax

Code :

(define function (myFunction{x , y} := (x * y)))

Function-calling command :

(print myFunction{2 , 3})

In this example , AI print number 6 .

The special feature be that human user already more detailedly group element in function definition more precisely specify element relationship .

This make it easier and quicker for AI system interpret human user intention .

AI system must still use regular expression recognize the inner bracket {x , y} .

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(2.21 . PHAM EXECUTING ORDER OF ELEMENT

Define Pham executing order of element aa :

Given element aa in Pham language codepage .

( if (there be element AA of strict form (AA{1} and AA{2} and .. AA{n}) contain aa at level-1 of hierarchy structure of AA) then ( ((Pham executing order) of aa) := ((Pham executing order) of AA{1}) ) else ( ((Pham executing order) of aa) := ((logical executing order) of aa) ) )

Note : Pham executing order of aa be always higher than Pham executing order of AA .

Thus by Pham executing order principle (in next part) , AA be always in examination before interpret aa .

Pham language codepage have smallest Pham executing order .

Explanation :

We know that each element have unique logical executing order .

But Pham language allow multiple element share same Pham executing order .

This happen when human user group element use word and .

This rule turn separate element into 1 single execution unit .

It allow AI system understand which matter happen at same time .

Note on well-definedness : AA{1} be element in Pham language codepage . Every element in Pham language codepage always have one unique Pham index and thus one unique logical executing order . Thus Pham executing order of AA{1} be always well-defined . This guarantee the definition of Pham executing order of aa be always non-circular and always resolve .

Example : Code :

(AA := ( (task 1) and (task 2) and (task 3) ))

In this example , (task 1) , (task 2) , and (task 3) all have same Pham executing order .

(2.21.1 . Avoid notation mistake for Pham executing order and parallel executing order

Declaration of Qualified Conjunction :

Pham executing order rule (where element inherit order of first element) apply only for conjunction which be make explicit :

(aa{1} and aa{2} .. and aa{n})

In this explicit conjunction , there be one strict rule .

Between any pair of adjacent word 'and' , there must be strictly only 1 Pham language element .

Also , before first word 'and' and after last word 'and' , there must be strictly only 1 Pham language element .

Note : Pham executing order of nested element aa be always big higher than Pham executing order of container AA . Thus , container AA be always examine before nested element aa be interpret .

Example of Qualified Conjunction :

Code :

(aa1 and aa2 and aa3)

This be qualified conjunction .

It have exactly 1 element between each 'and' .

It have exactly 1 element at begin and 1 element at end .

AI system apply Pham executing order rule on it .

All element inside conjunction share same parallel execution order .

Example of Not Qualified Conjunction :

Code 1 :

(aa1 and aa2 aa3 and aa4)

Code 2 :

(aa1 kk and aa2 and aa3)

Code 3 :

(aa1 and aa2 and aa3 qq)

In this example , this 3 element all be not qualified conjunction .

AI system not apply parallel executing order rule on them .

The reason be : there be more than 1 Pham language element between pair of adjacent 'and' (like aa2 aa3) , or there be more than 1 element at begin (like aa1 kk) , or there be more than 1 element at end (like aa3 qq) .

Instead , AI system use logical executing order (Pham index) for each element inside .

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(2.22 . PHAM FLAT ORDER OF ELEMENT

In Pham language codepage , Pham language codepage itself have smallest logical executing order .

Thus by convention , reckon Pham flat order of Pham language codepage 0 .

After Pham language codepage , all next element with 1 same smallest Pham executing order will have Pham flat order as 1 .

After all element with Pham flat order as 1 , next element with 1 same smallest Pham executing order will have Pham flat order as 2 . And so on .. .

Thus , in Pham language codepage , each element have Pham flat order as 1 integer .

In Pham language codepage , many element can have 1 same Pham flat order .

Explanation :

Pham flat order group element into horizontal layer of execution .

If element have same Pham flat order , they belong same execution step .

This structure allow AI system organize complex parallel task into simple numbered sequence of step .

Example : Code :

( (clean house) ( (wash dish) and (cook soup) ) (eat food) )

(clean house) have Pham flat order 1 .

(wash dish) and (cook soup) both have Pham flat order 2 .

(eat food) have Pham flat order 3 .

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(2.23 . THE PHAM EXECUTING ORDER PRINCIPLE

Declaration of the Executing Order :

In the Pham language codepage , the Pham executing order define the exact physical execution and interpretation sequence of element by the AI system .

Equivalent form : In the Pham language codepage , the Pham flat order define the exact physical execution and interpretation sequence of element by the AI system .

Principle 1 : (Sequential Execution) : Element aa with a smaller Pham executing order be execute first .

Principle 2 : (Parallel Execution) : Element bb with the same Pham executing order be execute simultaneously in parallel . But If the AI system cannot perform parallel execution/interpretation of bb , then the AI system can follow logical executing order of bb to serially execute/interpret bb .

The AI Workflow Interpretation :

Because the outermost Pham language codepage have the smallest executing order (Order 0) , the AI system will always interpret it first .

The workflow proceed as follow :

The AI system select an element and attempt interpret it .

Once the AI system finish interpret the element (or determine it not know how interpret it) , it immediately switch the next element with the next smallest Pham executing order .

This rule guarantee that the AI system always work in a strict logical sequence while maximize parallel processing power whenever possible .

This directly minimize AI energy consumption by ensure the full , efficient utilization of hardware thread .

Code Example Code :

( ((A := 10) and (B := 20)) (print (A + B)) )

Order 0 : The AI system first see the entire codepage or the outermost parent group .

Order 1 : The AI system execute (A := 10) and (B := 20) at the exact same time use parallel hardware thread .

Order 2 : The AI system execute (print (A + B)) last , because its logical order depend on the previous calculation be complete .

(2.23.1 . COROLLARY : Pham language have ABSOLUTE LOGICAL ABSTRACTNESS

Definition of Absolute Logical Abstractness : A language possess absolute abstractness if it can define a command (CMD) once , and then allow the system interpret that command in an infinite number of vastly different way depend on the context .

The language never need redefine or modify the original defined command CMD , and the overall logic flow of the system be always safely preserve .

The Formal Proof :

Let us examine an arbitrary command , cmd , in the Pham language .

In the Pham language , this command cmd be always a single element .

This element cmd can exist as a nested element inside a larger containing element , CMD .

According the Pham executing order principle , an outer containing element always have a smaller Pham executing order than its nested inner element .

Therefore , the AI system will always interpret the container CMD before it interpret the nested cmd .

Because the container be interpret first , it establish the physical context and dictate how cmd should be interpret later .

This process never require modify the original definition of cmd , nor it break the system logic .

If cmd be nest inside CMD1 , then CMD1 dictate the interpretation rule .

If cmd be nest inside CMD2 , then CMD2 dictate a different set of rule .

The core element cmd remain beautifully untouched .

Example of Absolute Logical Abstractness of Pham language :

Consider the following baseline command , cmd :

( cmd := (print Hello (3 time)) )

Example 1 (Physical Execution) : Code :

(in (main window) cmd)

Here , the containing element CMD1 be (in (main window) cmd) .

Because the AI observe this container first , it execute cmd by physically print the word "Hello" 3 time inside the main window .

Example 2 (Abstract Execution) : Code :

((in (main window) cmd) (this pham language message be instruction for AI write Python code))

Here , the containing element CMD2 be :

((in (main window) cmd) (this pham language message be instruction for AI write Python code))

The containing element CMD2 wrap the entire instruction .

Because CMD2 have the smallest executing order , the AI system observe it firstly .

This new container alter the reality of the execution : the AI interpret cmd by provide the text content of the required Python code .

There be no physical print of "Hello" in the window for the cmd .

The interpretation be simply the output text .

This example perfectly demonstrate the absolute logical abstractness of Pham language .

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(2.24 . PHAM GENERALIZED BOOLEAN AXIOM

Pham generalized Boolean axiom serve as fundamental basic apply boolean-liked operator on pham language element .

(2.24.1 . Pham generalized Boolean basic axiom :

Given (a b c d) be 4 arbitrary Pham language element .

Given (TRUE FALSE) be 2 standard Boolean value .

(2.24.1.1 Pham generalized Boolean Implication axiom :

((a -> b) =! ((not a) or b))

Example :

((rain -> wet) =! ((not rain) or wet))

In this example : meaning : If it rain , then it be wet .

This be same as : it be not rain or it be wet .

)

(2.24.1.2 Pham Idempotence of Implication axiom :

(((a -> b) and (a -> b)) =! (a -> b))

Example :

(((fire -> hot) and (fire -> hot)) =! (fire -> hot))

In this example : meaning : Repeat same logical implication not change truth state .

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(2.24.1.3 Pham selection interpretation axiom :

(((a or b) -> a) must be in (TRUE , FALSE))

(((a or b) -> b) must be in (TRUE , FALSE))

Interpretation : ((a or b) -> a) be always (Boolean TRUE) or (Boolean FALSE) .

Example :

(((apple or orange) -> apple) must be in (TRUE , FALSE))

In this example : meaning : Selection of apple from choice between apple or orange must result in 1 Boolean outcome .

)

(2.24.1.4 Pham TRUE-AND preservation axiom :

((TRUE and a) =! a)

((a and TRUE) =! a)

Interpretation : AI system can simplify explicit conjunction (TRUE and a) to a .

Example :

((TRUE and (print Hello)) =! (print Hello))

In this example : meaning : Confirm truth not change execution of command .

)

(2.24.1.5 Pham FALSE-AND annihilation axiom :

((FALSE and a) =! FALSE)

((a and FALSE) =! FALSE)

Interpretation in final physical stage : ignore interpret a . Take outcome of (FALSE and a) as boolean FALSE value . The AI system can simplify explicit (FALSE and a) to FALSE .

Example :

((FALSE and (calculate infinite sequence)) =! FALSE)

In this example : Meaning : If first condition be FALSE , AI system immediately stop .

It not execute heavy task . This save massive energy .

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(2.24.1.6 Pham FALSE-OR preservation axiom :

((FALSE or a) =! a)

((a or FALSE) =! a)

Interpretation : AI system can simplify explicit (FALSE or a)  to a .

Example :

((FALSE or (open door)) =! (open door))

In this example : meaning : FALSE state not block execution of matter in disjunction .

)

(2.24.1.7 Pham TRUE-OR annihilation axiom :

((a or TRUE) =! TRUE)

((TRUE or a) =! TRUE)

Interpretation : TRUE be dominant champion choice in (physical selection action) (a or TRUE) .

The AI system can simplify explicit (TRUE or a) to boolean TRUE .

The AI system can even ignore examine a in (TRUE or a) .

The AI system can simply ignore all other choice to take outcome of (a or TRUE) as TRUE .

Example :

(((buy milk) or TRUE) =! TRUE)

In this example : meaning : AI system can simply take outcome of ((buy milk) or TRUE) as TRUE , despite all other choice , and of course not need care about (buy milk) .

)

(2.24.1.8 Pham generalized De Morgan axiom :

((not (a and b)) =! ((not a) or (not b)))

((not (a or b)) =! ((not a) and (not b)))

Example :

((not(hot and dry)) =! ((not hot) or (not dry)))

)

(2.24.1.9 Pham negation annihilation axiom :

((a and (not a)) =! FALSE)

Interpretation : AI system can simplify (a and (not a)) to Boolean FALSE . AI system can immediately ignore examine a , (not a) in the element (a and (not a)) to save energy .

((not (not TRUE)) =! TRUE)

((not (not FALSE)) =! FALSE)

How about (not (not a)) ?

Pham language not specify any concrete outcome for ((not (not A)) == A) , where A be not Boolean value .

The AI system must use AI general knowledge and aware context to infer outcome for (not (not A)) .

Pham axiom not specify any concrete outcome for ((a or (not a)) == TRUE) .

The AI system use AI general knowledge and aware context infer outcome for (a or (not a)) .

In general , (if (a is not Boolean variable) then (outcome of (a or (not a)) be often not define as Boolean TRUE)) . Thus the AI system must not aggressively assign (a or (not a)) to TRUE when a is not Boolean value . Again : Pham language not specify any specific outcome for (a or (not a)) .

)

(2.24.1.10 Pham Commutation of AND axiom :

((a and b) =! (b and a))

Example :

((sun and moon) =! (moon and sun))

Meaning : Order of conjunction not change physical essence of matter .

Because in (aa and bb) , the element aa and element bb have same Pham executing order , thus they be examine in parallel , thus there should not differ which one be first .

)

)

(2.24.2 . Pham Concrete Selection Axiom :

(2.24.2.1 Axiom :

Given (a b) be 2 Pham language element .

((if ((a or b) -> a) then ((a or b) =! a)))

((if((a or b) -> b) then ((a or b) =! b)))

Equivalent alternative form :

(((a or b) -> a) =! ((a or b) == a))

(((a or b) -> b) =! ((a or b) == b))

Example :

((if ((left or right) -> left) then ((left or right) =! left)))

In this case : meaning : If selection trigger point left , then whole disjunction become strictly left .

)

(2.24.2.2 Corollary of Pham Selection Exclusion axiom :

((if (((a or b) -> a) and ((a or b) -> b)) then (a =! b)))

Equivalent alternative form of Pham Selection Exclusion Corollary :

((((a or b) -> a) and ((a or b) -> b)) =! (a == b))

Example :

((if (((red or blue) -> red) and ((red or blue) -> blue)) then (red =! blue)))

In this example : Meaning : If AI select both at same time , they must be same identical matter .

)

)

(2.24.3 . Pham Completeness of Selection Axiom :

Given (a b) be 2 Pham language element .

((((a or b) -> a) or ((a or b) -> b)) =! TRUE)

Example :

((((work or sleep) -> work) or ((work or sleep) -> sleep)) =! TRUE)

In this example : Meaning : AI system be force choose at least 1 outcome . Choice not can be empty .

)

)

(2.25 . PHAM UNDETERMINISTIC SELECTION THEOREM

(proof be in annex 1)

Pham undeterministic selection theorem be backbone of Pham language 202602 .

It be fundamental bridge between ((AI interpreting function mapping) of (partly-deterministic argument selection)) and ((partly-deterministic selection) of (AI interpreting function mapping of absolute-deterministic argument selection)) .

Pham undeterministic selection theorem allow transform original Pham language message with undeterministic argument selection into full-info absolute equivalent undeterministic selection of Pham language message branch with deterministic argument .

This transformation not leak info of original Pham language message .

This transformation fully transfer info of original Pham language message .

Thus it be extreme useful in robotic , automation , clean hydrogen energy , nuclear energy safety , and other sector .

(2.25.1 . Pham Undeterministic Selection Theorem for Simple Case

( (a{1} .. a{k-1} (a{k} or a{k+1}) a{k+2} .. a{n}) =! ( (((a{k} or a{k+1}) -> a{k}) and (a{1} .. a{k-1} a{k} a{k+2} .. a{n})) or (((a{k} or a{k+1}) -> a{k+1}) and (a{1} .. a{k-1} a{k+1} a{k+2} .. a{n})) ) )

Real Life Meaning : If 1 part of 1 system exist in state of choice , then the whole system exist as 1 choice between 2 distinct physical outcome .

Real Life Application : Used in Emergency Automation . If 1 sensor detect (fire or flood) , the system logic must immediately expand into 2 deterministic safety path .

)

(5 Meaningful Deep Nested Example (Simple Case) :

Example :

( (((hydrogen cell) (state (active or standby))) and ((coolant valve) (mode (open or shut)))) =! ( ( ((active or standby) -> active) and ((open or shut) -> open) and ((hydrogen cell) (state active) and ((coolant valve) (mode open))) ) or ( ((active or standby) -> standby) and ((open or shut) -> shut) and ((hydrogen cell) (state standby) and ((coolant valve) (mode temporary shutdown))) ) ) )

Example :

( (((space rocket) (fuel (liquid or gas))) and ((engine stage) (ignition (yes or no)))) =! ( ( ((liquid or gas) -> liquid) and ((yes or no) -> yes) and ((space rocket) (fuel liquid) and ((engine stage) (ignition yes))) ) or ( ((liquid or gas) -> gas) and ((yes or no) -> no) and ((space rocket) (fuel gas) and ((engine stage) (ignition no))) ) ) )

Example :

( (((industrial robot) (arm position (left or right))) and ((gripper state) (grip (on or off)))) =! ( ( ((left or right) -> left) and ((on or off) -> on) and ((industrial robot) (arm position left) and ((gripper state) (grip on))) ) or ( ((left or right) -> right) and ((on or off) -> off) and ((industrial robot) (arm position right) and ((gripper state) (grip off))) ) ) )

Example :

( (((medical scan) (image type (mri or xray))) and ((doctor report) (status (ready or pending)))) =! ( ( ((mri or xray) -> mri) and ((ready or pending) -> ready) and ((medical scan) (image type mri) and ((doctor report) (status ready))) ) or ( ((mri or xray) -> xray) and ((ready or pending) -> pending) and ((medical scan) (image type xray) and ((doctor report) (status pending))) ) ) )

Example :

( (((global network) (data protocol (tcp or udp))) and ((encryption level) (type (high or low)))) =! ( ( ((tcp or udp) -> tcp) and ((high or low) -> high) and ((global network) (data protocol tcp) and ((encryption level) (type high))) ) or ( ((tcp or udp) -> udp) and ((high or low) -> low) and ((global network) (data protocol udp) and ((encryption level) (type low))) ) ) )

)

(2.25.2 . Pham undeterministic selection theorem for synchronized variable case

Given (f x) be pham language element value function f of variable (pham language element x) .

Given (g x) be pham language element value function g of variable (pham language element x) .

Formula 1 :

( (f (g (a or b))) =! ( (((a or b) -> a) and (f (g a))) or (((a or b) -> b) and (f (g b))) ) )

Formula 2 :

( (f (g (a or b))) =! (f ( (((a or b) -> a) and (g a)) or (((a or b) -> b) and (g b)) ) ) )

Real Life Meaning : Choosing the final output of 1 process chain be identical choosing the input element at the start of that chain .

Real Life Application : Used in Hydrogen Fuel Storage . Calculating the risk of choose (tank_A or tank_B) be identical choose between the risk of tank_A and the risk of tank_B .

Meaningful Deep Nested Example (Formula 1) :

Example :

( ((notify (message format (type (text or html))))) =! ( ( (((text or html) -> text) and (notify (message format text))) or (((text or html) -> html) and (notify (message format html))) ) ) )

Example :

( ((encrypt (secure hash (data (file or folder))))) =! ( ( (((file or folder) -> file) and (encrypt (secure hash file))) or (((file or folder) -> folder) and (encrypt (secure hash folder))) ) ) )

Example :

( ((speak (ai translate (word (hello or ciao))))) =! ( ( (((hello or ciao) -> hello) and (speak (ai translate hello))) or (((hello or ciao) -> ciao) and (speak (ai translate ciao))) ) ) )

Example :

( ((calculate (math log (value (10 or 100))))) =! ( ( (((10 or 100) -> 10) and (calculate (math log 10))) or (((10 or 100) -> 100) and (calculate (math log 100))) ) ) )

Example :

( ((execute (system script (mode (safe or fast))))) =! ( ( (((safe or fast) -> safe) and (execute (system script safe))) or (((safe or fast) -> fast) and (execute (system script fast))) ) ) )

Meaningful Deep Nested Example (Formula 2) :

Example :

( ((analyze (biolab sample (dna (plant or animal))))) =! (analyze (biolab sample ( (((plant or animal) -> plant) and (dna plant)) or (((plant or animal) -> animal) and (dna animal)) ) ) ) )

Example :

( ((render (gpu engine (texture (wood or stone))))) =! (render (gpu engine ( (((wood or stone) -> wood) and (texture wood)) or (((wood or stone) -> stone) and (texture stone)) ) ) ) )

Example :

( ((synthesize (audio wave (tone (sine or square))))) =! (synthesize (audio wave ( (((sine or square) -> sine) and (tone sine)) or (((sine or square) -> square) and (tone square)) ) ) ) )

Example :

( ((predict (weather model (trend (up or down))))) =! (predict (weather model ( (((up or down) -> up) and (trend up)) or (((up or down) -> down) and (trend down)) ) ) ) )

Example :

( ((authorize (security hub (lock (open or shut))))) =! (authorize (security hub ( (((open or shut) -> open) and (lock open)) or (((open or shut) -> shut) and (lock shut)) ) ) ) )

)

(2.25.3 . Pham undeterministic selection theorem for asynchronized variable case

Given (f x y z) be pham language element value function f of 3 variable ((pham language element x) (pham language element y) (pham language element z)) .

Given (g x y z) be pham language element value function g of 3 variable ((pham language element x) (pham language element y) (pham language element z)) .

Formula 3 :

( (f (g (a or b) a b) a b) =! ( (((a or b) -> a) and (f (g a a b) a b)) or (((a or b) -> b) and (f (g b a b) a b)) ) )

Formula 4 :

( (f (g (a or b) a b) a b) =! (f ( (((a or b) -> a) and (g a a b)) or (((a or b) -> b) and (g b a b)) ) a b) )

Real Life Meaning : 1 choice (a or b) can be resolve while keep the element a and b as fixed context elsewhere in the same command .

Real Life Application : Used in Industrial Automation . If 1 robot select between (Tool_1 or Tool_2) , it keep the specification of both Tool_1 and Tool_2 in its local memory ensure 1 safe transition .

Meaningful Deep Nested Example (Formula 3) :

Example :

( (rank (match result (win or loss) win loss) win loss) =! ( ( (((win or loss) -> win) and (rank (match result win win loss) win loss)) or (((win or loss) -> loss) and (rank (match result loss win loss) win loss)) ) ) )

Example :

( (verify (user login (id_1 or id_2) id_1 id_2) id_1 id_2) =! ( ( (((id_1 or id_2) -> id_1) and (verify (user login id_1 id_1 id_2) id_1 id_2)) or (((id_1 or id_2) -> id_2) and (verify (user login id_2 id_1 id_2) id_1 id_2)) ) ) )

Example :

( (move (path coord (X or Y) X Y) X Y) =! ( ( (((X or Y) -> X) and (move (path coord X X Y) X Y)) or (((X or Y) -> Y) and (move (path coord Y X Y) X Y)) ) ) )

Example :

( (adjust (temp level (hot or cold) hot cold) hot cold) =! ( ( (((hot or cold) -> hot) and (adjust (temp level hot hot cold) hot cold)) or (((hot or cold) -> cold) and (adjust (temp level cold hot cold) hot cold)) ) ) )

Example :

( (track (object shape (ball or cube) ball cube) ball cube) =! ( ( (((ball or cube) -> ball) and (track (object shape ball ball cube) ball cube)) or (((ball or cube) -> cube) and (track (object shape cube ball ball cube) ball cube)) ) ) )

Meaningful Deep Nested Example (Formula 4) :

Example :

( (simulate (price trend (buy or sell) buy sell) buy sell) =! (simulate ( ( (((buy or sell) -> buy) and (price trend buy buy sell) buy sell) ) or ( (((buy or sell) -> sell) and (price trend sell buy sell) buy sell) ) ) buy sell ) )

Example :

( (analyze (dna strand (type_A or type_B) type_A type_B) type_A type_B) =! (analyze ( ( (((type_A or type_B) -> type_A) and (dna strand type_A type_A type_B) type_A type_B) ) or ( (((type_A or type_B) -> type_B) and (dna strand type_B type_A type_B) type_A type_B) ) ) type_A type_B ) )

Example :

( (dispatch (task priority (low or high) low high) low high) =! (dispatch ( ( (((low or high) -> low) and (task priority low low high) low high) ) or ( (((low or high) -> high) and (task priority high low high) low high) ) ) low high ) )

Example :

( (filter (signal frequency (fast or slow) fast slow) fast slow) =! (filter ( ( (((fast or slow) -> fast) and (signal frequency fast fast slow) fast slow) ) or ( (((fast or slow) -> slow) and (signal frequency slow fast slow) fast slow) ) ) fast slow ) )

Example :

( (create (model mesh (v1 or v2) v1 v2) v1 v2) =! (create ( ( (((v1 or v2) -> v1) and (model mesh v1 v1 v2) v1 v2) ) or ( (((v1 or v2) -> v2) and (model mesh v2 v1 v2) v1 v2) ) ) v1 v2 ) )

)

(2.25.4 . Commutation feature of Pham undeterministic selection theorem :

(proof in annex 1)

Pham language codepage can contain many disjunctive pair (ai or bi) in different element .

Apply (Pham undeterministic selection theorem) on disjunctive pair (ai or bi) and disjunctive pair (aj or bj) be commutative .

( If (there be not dedicated purpose) then (should follow (Pham executing order principle) by (Pham executing order of element criterion) when (apply (pham undeterministic selection theorem) on disjunctive pair (ai or bi) )) )

REAL LIFE MEANING :

This feature mean that in complex system with many choice , the final list of certain outcome not depend on which choice you look at firstly .

It guarantee that AI system not get different result just because it apply (Pham undeterministic selection theorem) on a disjunctive pair (a{i} or b{j}) before it apply (Pham undeterministic selection theorem) on another disjunctive pair (a{j} or b{j}) .

REAL LIFE APPLICATION :

Used in Automated Traffic Control .

A system must manage (Light_A (Red or Green)) and (Light_B (Red or Green)) .

The Commutation feature ensure that the final safe state of the intersection be the same whether the AI check Light_A or Light_B firstly .

This prevent accident and save energy .

Pham language allow many choice (or) in 1 message .

This feature prove that the order of solve this choice not change the final set of certain result .

This be Commutation .

By default , AI system solve choice from left right match the Pham executing order of element .

If a special reason exist , AI system can solve them in a different order . The result will be same , thank to commutation feature of Pham undeterministic selection theorem .

ILLUSTRATING MEANINGFUL DEEP-NESTED EXAMPLE :

Example 1 :

(Hydrogen Safety Monitor)

( ( ((monitor cell) (temp (high or low))) and ((alert signal) (type (vocal or visual))) ) =! ( ( (((high or low) -> high) and (((vocal or visual) -> vocal) and ((monitor cell) (temp high) and (alert signal (type vocal))))) or (((high or low) -> high) and (((vocal or visual) -> visual) and ((monitor cell) (temp high) and (alert signal (type visual))))) or (((high or low) -> low) and (((vocal or visual) -> vocal) and ((monitor cell) (temp low) and (alert signal (type vocal))))) or (((high or low) -> low) and (((vocal or visual) -> visual) and ((monitor cell) (temp low) and (alert signal (type visual))))) ) ) )

Example 2 :

(Robotic Surgery Logic)

( ( ((surgical robot) (tool (laser or needle))) and ((target tissue) (depth (surface or deep))) ) =! ( ( (((laser or needle) -> laser) and (((surface or deep) -> surface) and ((surgical robot) (tool laser) and (target tissue (depth surface))))) or (((laser or needle) -> laser) and (((surface or deep) -> deep) and ((surgical robot) (tool laser) and (target tissue (depth deep))))) or (((laser or needle) -> needle) and (((surface or deep) -> surface) and ((surgical robot) (tool needle) and (target tissue (depth surface))))) or (((laser or needle) -> needle) and (((surface or deep) -> deep) and ((surgical robot) (tool needle) and (target tissue (depth deep))))) ) ) )

Example 3 :

(Space Flight Path)

( ( ((thruster stage) (power (max or min))) and ((navigation mode) (target (moon or mars))) ) =! ( ( (((max or min) -> max) and (((moon or mars) -> moon) and ((thruster stage) (power max) and (navigation mode (target moon))))) or (((max or min) -> max) and (((moon or mars) -> mars) and ((thruster stage) (power max) and (navigation mode (target mars))))) or (((max or min) -> min) and (((moon or mars) -> moon) and ((thruster stage) (power min) and (navigation mode (target moon))))) or (((max or min) -> min) and (((moon or mars) -> mars) and ((thruster stage) (power min) and (navigation mode (target mars))))) ) ) )

)

(2.25.5 . PHAM UNDETERMINISTIC SELECTION THEOREM FOR MULTI-ELEMENT DISJUNCTION :

Multi-element disjunction :

(aa{1} or aa{2} .. or aa{n})

(Pham undeterministic selection theorem for multi-element disjunction) be absolute analogous (Pham undeterministic selection theorem for 2-element disjunction) .

EXPLANATION :

This theorem allow AI system resolve 1 nested choice contain many possible part .

It transform 1 complex message into a selection of many certain branch .

Each branch represent exactly 1 deterministic outcome .

This prevent AI system from use "superposition" logic which waste energy .

REAL LIFE MEANING :

In a complex system , 1 decision point can have many path .

This theorem mean the system can split into many individual version of itself .

Each version handle 1 specific choice .

This turn a messy choice into 1 set of clear , certain action .

ILLUSTRATING MEANINGFUL DEEP-NESTED EXAMPLE :

Example 1 :

(Advanced Robotic Sorter)

( ( (sort ((package item) (material (metal or plastic or glass)))) ) =! ( ( (((metal or plastic or glass) -> metal) and (sort ((package item) (material metal)))) ) or ( (((metal or plastic or glass) -> plastic) and (sort ((package item) (material plastic)))) ) or ( (((metal or plastic or glass) -> glass) and (sort ((package item) (material glass)))) ) ) )

Example 2 :

(Hydrogen Plant Control)

( ( (execute ((hydrogen plant) (emergency mode (vent or flush or seal)))) ) =! ( ( (((vent or flush or seal) -> vent) and (execute ((hydrogen plant) (emergency mode vent)))) ) or ( (((vent or flush or seal) -> flush) and (execute ((hydrogen plant) (emergency mode flush)))) ) or ( (((vent or flush or seal) -> seal) and (execute ((hydrogen plant) (emergency mode seal)))) ) ) )

Pham language allow 1 choice with many part (aa{1} or aa{2} .. or aa{n}) .

This theorem prove that 1 message with many choice be same as 1 list of certain result .

Each certain result use only 1 part of the choice .

AI system use this solve complex problem by look at every possible certain path .

)

(2.25.6 . AVOID NOTATION MISTAKE FOR PHAM UNDETERMINISTIC SELECTION THEOREM

Declaration of Qualified Disjunction :

Pham undeterministic selection theorem apply only for disjunction which be form explicit :

(aa{1} or aa{2} .. or aa{n})

In this explicit disjunction , there be one strict rule .

Between any pair of adjacent word 'or' , there must be strictly only 1 Pham language element .

Also , before first word 'or' and after last word 'or' , there must be strictly only 1 Pham language element .

Example of Qualified Disjunction Code :

(aa or bb or cc or dd)

This be qualified disjunction .

It have exactly 1 element between each 'or' .

It have exactly 1 element at begin and 1 element at end .

AI system apply Pham undeterministic selection theorem on it .

Example of Not Qualified Disjunction :

Code 1 :

(aa or bb or cc pp or dd)

Code 2 :

(aa kk or bb or cc or dd)

Code 3 :

(aa or bb or cc or dd qq)

In this example , this 3 element all be not qualified disjunction .

AI system not apply Pham undeterministic selection theorem on them .

The reason be : there be more than 1 Pham language element between pair of adjacent 'or' (like cc pp) , or there be more than 1 element at begin (like aa kk) , or there be more than 1 element at end (like dd qq) .

)))

(3 . PHAM LANGUAGE TRANSLATION ALGORITHM ( ADVANCED PART ) :

(Content of this part be in Pham language text .)

(Term correctness convention :

(AI must always call correct dedicated term of each concept .)

(AI should avoid call weird short ambiguous term of concept .)

)

((Pham language translation algorithm) help translate intention into (Pham language message) .)

((Pham language translation algorithm) also help translate natural language text into (Pham language message) .)

(Pham language translation algorithm with Pham language can truly lead AI actually think instead of AI simply mechanically list word in prioritized probabilistic ordered sequence as 2026y today .)

(3.1 . Pham Conjuncture Operator :

(Pham language 202602 originally be design with its integrated Pham conjuncture operator .)

(Pham conjuncture operator be a very powerful tool in Pham language .)

(However , Pham conjuncture operator require user must very precisely use it avoid ambiguity because Pham conjuncture operator can easily lead to hidden , very small , but subtle inaccuracy .)

(By that reason , Pham language 202602 not generally include Pham conjuncture operator in core human syntax .)

(But , Pham conjuncture operator can be extremely useful resolve subtle uncertainty of natural language text .)

(Thus , in this section , Pham conjuncture operator be use dedicatedly only by AI system to translate natural language text into Pham language text .)

(Definition of Pham Conjuncture Operator :

(Pham conjuncture operator introduce a new operator with new conjuncture symbol : and!)

(Given a{i} be any Pham language element)

(Pham conjuncture operator mathematically state :)

( (a{1} a{2} .. a{k-1} (a{k} and! a{k+1}) a{k+2} .. a{k+n}) =! ((a{1} a{2} .. a{k-1} a{k} a{k+2} .. a{k+n}) and! (a{1} a{2} .. a{k-1} a{k+1} a{k+2} .. a{k+n})) )

(Apply Pham conjuncture operator only and only locally .)

)

(Example of Pham Conjuncture Operator :

(Given (a b c d e f g) be 7 independent Pham language element)

(Pham conjuncture operator state exactly this result :)

( ((a and! b) c) =! ((a c) and! (b c)) )

( (a (b and! c)) =! ((a b) and! (a c)) )

( (a (b and! c) d) =! ((a b d) and! (a c d)) )

( ((a and! b and! c) d) =! ((a d) and! (b d) and! (c d)) )

Note : This example show expansion over many element .

( (a b (c (d and! e) f) g) =! (a b ((c d f) and! (c e f)) g) )

Note : This example show precise local scope expansion for deep-nested element .

Apply Pham conjuncture operator only locally .

)

(Important note on execution :

(Pham conjuncture operator be not commutative with Pham undeterministic selection theorem .)

(Thus , translation process require explicitly commit to Pham executing order principle when AI system apply Pham conjuncture operator .)

)

)

(3.2 . Pham Language Grammar Function :

(Pham language grammar function be to help create Pham language text .)

(AI system can store much many Pham language grammar function .)

(Local system can also store much many (Pham language grammar function))

(Required feature of Pham language grammar function :

(A Pham language grammar function can accept arbitrary number of input parameter .)

(Any input parameter of Pham language grammar function must always be strictly 1 whole qualified Pham language element .)

(And , return value of Pham language grammar function must always be strictly 1 whole qualified Pham language element .)

)

(Specific feature of Pham language grammar function :

(Function name of Pham language grammar function should be meaningful Pham language element (Pham language atomic element or Pham language compound element ))

(Because Pham language 202602 allow use (Pham language compound element) as function name , thus AI system possible flexibly and meaningfully choose function name)

(In declaration of (Pham language grammar function) , allow each input argument as 1 whole compound pham language element too)

)

(Important note : Return value not can be 2 or more Pham language element .

)

(Pham language grammar function are dedicatedly for AI system)

(AI system can (create and use) Pham language grammar function for (Pham indexed grammar function repository))

((Human user usually not need directly use Pham language grammar function)

thus (long meaningful (function name) of (Pham language grammar function) be ((not cause hassle for human user) and (to lead rich meaningful info (function name) for AI system))))

(3.2.1 . Create Pham language grammar function from basic built-in pham language command :

((Each basic pham language command is always a 1 whole qualified pham language element)

(Thus each pham language command often hint pattern to create text string as (qualified returned value) of Pham language grammar function)

(For example : Pham language command

(if (condition) then (statement{1} statement{2} .. statement{n})) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : (((if-clause) or (conditional execution) or (conditional branching)) (general form 1)))

(function input parameter : (the_condition) (the_execution))

(function implementation : (return text string (if (the_condition) then (the_execution))))))

How to call the function ? :

((((if-clause) or (conditional execution) or (conditional branching)) (general form 1)) (my condition) (my execution))

It will return :

((if ((my condition)) then ((my execution))) as (a text string))

)

(For example : Pham language command

(if (condition) then (statement{1} statement{2} .. statement{n})) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : (((if-clause) or (conditional execution) or (conditional branching)) (general form 2)))

(function input parameter : (the_condition) (the_execution))

(function implementation : (return text string (if the_condition then the_execution)))))

How to call the function ? :

((((if-clause) or (conditional execution) or (conditional branching)) (general form 2)) (my condition) (my execution))

It will return :

((if (my condition) then (my execution)) as (a text string))

)

(For example : Pham language command (the binary grouping command) :

(aa bb) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : ((grouping and (binary grouping))(general form)))

(function input parameter : (first_argument : arbitrary pham language element) ((second argument) : arbitrary pham language element))

(function implementation : (return (text string (first_argument (second argument)))))))

How to call the function ? :

(((grouping and (binary grouping))(general form)) aaa bbb)

It will return :

(aaa bbb)

)

(For example : Pham language command

(for (item in (some set)) do (subCommand_1 .. subCommand_n)) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : (((loop) or (loop command)) (for item in set form)))

(function input parameter : the_item the_set the_execution)

(function implementation : (return text string (for (the_item in the_set) do (the_execution))))))

How to call the function ? :

((((loop) or (loop command)) (for item in set form)) (my item) (my set) (my execution))

It will return :

((for ((my item) in (my set)) do (my execution)) as (a text string))

)

(For example : Pham language command

((interested stuff) := (target stuff)) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : (((assignment) or (assignment command)) (standard form)))

(function input parameter : the_interested_stuff the_target_stuff)

(function implementation : (return text string (the_interested_stuff := the_target_stuff)))))

How to call the function ? :

((((assignment) or (assignment command)) (standard form)) (my stuff) (my target))

It will return :

(((my stuff) := (my target)) as (a text string))

)

(For example : Pham language command

((stuff aa) =! (stuff bb)) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : (((Pham-equality-knowledge) or (Pham-equality-knowledge command)) (standard form)))

(function input parameter : stuff_A stuff_B)

(function implementation : (return text string (stuff_A =! stuff_B)))))

How to call the function ? :

((((Pham-equality-knowledge) or (Pham-equality-knowledge command)) (standard form)) (my stuff) (my target))

It will return :

(((my stuff) =! (my target)) as (a text string))

)

(For example : Pham language command

((stuff aa) == (stuff bb)) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : (((boolean equality query) or (boolean query)) (standard form)))

(function input parameter : stuff_A stuff_B)

(function implementation : (return text string (stuff_A == stuff_B)))))

How to call the function ? :

((((boolean equality query) or (boolean query)) (standard form)) (my state) (my benchmark))

It will return :

(((my state) == (my benchmark)) as (a text string))

)

))

(3.2.2 . Create Pham language grammar function from formula of Pham axiom , Pham undeterministic selection theorem :

(Each Pham axiom and theorem is always a 1 whole qualified Pham language element .

Thus each axiom or theorem often hint pattern to create complex logical expansion as (qualified returned value) of Pham language grammar function .

(For example : Pham generalized Boolean Implication axiom

((a -> b) =! ((not a) or b)) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : (((boolean implication) or (implication expansion)) (standard form)))

(function input parameter : the_antecedent the_consequent)

(function implementation : (return text string ((the_antecedent -> the_consequent) =! ((not the_antecedent) or the_consequent))))))

How to call the function ? :

((((boolean implication) or (implication expansion)) (standard form)) (rain) (wet))

It will return :

(((rain -> wet) =! ((not rain) or wet)) as (a text string))

)

(For example : Pham undeterministic selection theorem (simple case)

((a{1} .. (a{k} or a{k+1}) .. a{n}) =! ((((a{k} or a{k+1}) -> a{k}) and (a{1} .. a{k} .. a{n})) or (((a{k} or a{k+1}) -> a{k+1}) and (a{1} .. a{k+1} .. a{n})))) .

It can hint pattern to create the following Pham language grammar function :

(define function (

(function name : (((undeterministic selection) or (choice expansion)) (simple form)))

(function input parameter : prefix the_choice_A the_choice_B suffix)

(function implementation : (return text string (

( (prefix (the_choice_A or the_choice_B) suffix) =!

( ( ((the_choice_A or the_choice_B) -> the_choice_A) and (prefix the_choice_A suffix) ) or

( ((the_choice_A or the_choice_B) -> the_choice_B) and (prefix the_choice_B suffix) ) )

)

)))))

How to call the function ? :

((((undeterministic selection) or (choice expansion)) (simple form)) (robot move) (left) (right) (to avoid obstacle))

It will return :

(( (robot move (left or right) to avoid obstacle) =! ( ( ((left or right) -> left) and (robot move left to avoid obstacle) ) or ( ((left or right) -> right) and (robot move right to avoid obstacle) ) ) ) as (a text string))

))

)

(3.2.3 . Create Pham language grammar function from natural language grammar phrase :

(In this specification , Pham language 202602 take grammar phrase of general standard English to explain Pham language grammar function concept )

(Each natural language have limited number of its used grammar phrase .

Number of all known grammar phrase of natural language of humanity is currently simply colossal tiny for the ability of any entry-level popular AI system today . Thus it is easy to load all known grammar phrase of all natural language to any entry-level AI system)

(Worldwide reality at 2026y today : All correct natural language sentence be write in form of known natural language grammar phrase .)

(Because a (natural language grammar phrase) is a stable structural unit of human communication , it hint a rigid geometric pattern for text construction .)

(Each (natural language grammar phrase) can hint pattern to generate corresponding various (Pham language grammar function))

(The AI system identify (natural language grammar phrase) in natural text and map it to a (qualified Pham language element) through the calling of (Pham language grammar function) .)

(For example :

English grammar phrase : subject verb object .

It hint generate Pham language grammar function :

(define function (

(function name : (((natural language) English) (grammar phrase) (general simple sentence)))

(function input parameter : theSubject theVerb theObject)

(function implementation : (return (text string (theSubject theVerb theObject))))))

How to call this function :

(((((natural language) English) (grammar phrase) (general simple sentence)) (subject verb object)) (all people) breath (depleted oxygen of air))

It will return :

((all people) breath (depleted oxygen of air))

)

(For example :

English grammar phrase : adjective adjective noun .

It hint generate Pham language grammar function :

(define function (

(function name : (((natural language) English) (grammar phrase) (adjective adjective noun)))

(function input parameter : (adjective 1) (adjective 2) noun)

(function implementation : (return (text string ((adjective 1) (adjective 2) noun))))))

How to call this function :

((((natural language) English) (grammar phrase) (adjective adjective noun)) yellow (bright flash) (new car))

It will return :

(yellow (bright flash) (new car))

)

(For example :

English grammar phrase : something in otherthing .

It hint generate Pham language grammar function :

(define function (

(function name : (((natural language) English) (grammar phrase) (something in something)))

(function input parameter : something1  (something 2))

(function implementation : (return (text string (something1 in (something 2)))))))

How to call this function :

((((natural language) English) (grammar phrase) (something in something)) (the cat eat) (black floor))

It will return :

((the cat eat) in (black floor))

)

(For example :

English grammar phrase : Who do what where .

It hint generate Pham language grammar function :

(define function (

(function name : (((natural language) English) (grammar phrase) (agent action location)))

(function input parameter : the_agent the_action the_location)

(function implementation : (return (text string ((the_agent) do (the_action) in (the_location)))))))

How to call this function :

((((natural language) English) (grammar phrase) (agent action location)) (a brave worker) (fix the pipe) (main factory))

It will return :

((a brave worker) do (fix the pipe) in (main factory))

)

(3.2.3.1 . Nesting Pham language grammar function make Pham language play as meta-language :

(natural language grammar phrase infer and generate Pham language grammar function)

(Nest these (Pham language grammar function) , in much many case , impact as **meta-language**)

(**Pham language playing as meta-language** allow AI system exactly identify the role of every word according to its (natural language grammar phrase) container .)

(Example :

(Given (phrase_adj_noun) be the (Pham language grammar function) of (grammar phrase (adjective noun)))

(Given (phrase_verb_in) be the (Pham language grammar function) of (grammar phrase (something verb in something)))

(A standard sentence : (The red apple stay in the wooden box))

(Step 1 : Identify internal groupings using (phrase_adj_noun) :)

(The (phrase_adj_noun red apple) stay in the (phrase_adj_noun wooden box))

(Intermediate result : (The (red apple) stay in the (wooden box)))

(Step 2 : Identify overall sentence structure using (phrase_verb_in) :)

(phrase_verb_in (The (phrase_adj_noun red apple)) stay (the (phrase_adj_noun wooden box)))

(Final translated result : ((The (red apple)) stay in (the (wooden box))))

(Interpretation : Instead of a flat string , the AI system see a hierarchy of (Pham language grammar function) calls . This meta-structure removes ambiguity by physically locking the relationship between word groups .)

(Example :

(Given (phrase_subject_verb_on_noun) be (the (Pham language grammar function) of (grammar phrase (subject verb on noun))))

(Given (phrase_subject_verb_object) be (the (Pham language grammar function) of (grammar phrase (subject verb object))))

(Step 1 : (the (Pham language grammar function) (phrase_subject_verb_on_noun)) generate the string result :)

((people waste on ocean) := (phrase_subject_verb_on_noun people waste ocean))

(Step 2 : Substitute (this result) into the argument of (the (Pham language grammar function) (phrase_subject_verb_object)) :)

(phrase_subject_verb_object (people waste on ocean) harm environment)

(Final translated result : ((people waste on ocean) harm environment))

)

(Example :

((the football command have good player and coach) (it also have robust discipline) and (it train well))

(Now put it in ((Pham language grammar function) of (grammar phrase (subject verb object))) return :

( ((the football command have good player and coach) (it also have robust discipline) and (it train well)) allow its championship title) )

)

(Interpretation : This process demonstrate how the (Pham language translation algorithm) use nested (Pham language grammar function) call to transform complex natural language intent into a strictly qualified (Pham language meta-message) .)

(Superiority of Expression : A message like ( ((the football command have good player and coach) (it also have robust discipline) and (it train well)) allow its championship title ) be nearly impossible to strictly express in standard English without adding ambiguous filler words or complex relative clauses . In (Pham language) , the (Pham language grammar function) physically lock the whole phrase ((the football command have good player and coach) (it also have robust discipline) and (it train well)) into a (single atomic subject) , allowing for a message that be both mathematically precise and structurally compact .)

)

)

)

)

(3.3 . Natural language legacy preservation requirement :

(AI model can have many sector-specific (Pham indexed grammar function repository) to explore sector-specific content)

((Pham language strongly emphasizment : arbitrary sector-specific (Pham indexed grammar function repository) must contain core set of (Pham language grammar function) , which are derived from all basic (natural language grammar phrase)) (It allow penetrate DNA of humanity natural language legacy into sector))

(Pham language to protect natural language legacy by (allow natural language in form of Pham language can be used in even very specific sector))

)

(3.4 . Pham indexed grammar function repository :

(Specified term : (Pham indexed grammar function repository))

(Pham indexed grammar function repository contain all Pham language grammar function)

(AI system can store many different (Pham indexed grammar function repository) for different application sector)

(In (Pham indexed grammar function repository) , function name of each Pham language grammar function be semantic meaningful compound pham language element)

(In (Pham indexed grammar function repository) , (function name) of each Pham language grammar function be indexed by worldwide accepted vocabulary for comfortable internal AI search engine)

(Because function name of Pham language grammar function can be compound pham language element , thus they be strictly hierarchy structure , thus they allow very drastically flexible indexing and categorizing by AI system internal search engine)

(Because input argument in declaration of Pham language grammar function can be compound pham language element , thus they be strictly hierarchy structure , thes they allow very drastically flexible indexing and categorizing by AI system internal search engine)

(Pham indexed grammar function repository be used in (Pham OO-translation algorithm) and (Pham Neighbor-Grouping Translation Algorithm))

( ( (Pham language , Pham language architecture , Pham undeterministic selection theorem , Pham translation algorithm be exclusively human intellectual effort of Pham Language Foundation . They be open-source and free for any use without relabelling)

but

(Each AI human company can have its own proprietary (Pham indexed grammar function repository)) )

(Pham language not specify any concrete info about specific (Pham indexed grammar function repository))

(Pham language Recommendation : (any specific (Pham indexed grammar function repository) must and also contain ((Pham language grammar function) , which is generated from all worldwide accepted popular (natural language grammar phrase))) (it will guarantee that the output Pham language text are always ((human friendly) and (to protect legacy humanity culture civilization))) ))

)

(3.5 . Pham OO-translation algorithm :

(Official specified concept term : (Pham OO-translation algorithm))

((Pham OO-translation algorithm) stand for the meaning (Pham object-oriented translation algorithm))

((Pham OO-translation algorithm) is use to translate (object-oriented abstract representation) into human friendly meaningful Pham language text)

((Pham OO-translation algorithm) use (((2 component) ((Pham indexed grammar function repository) and (AI internal self querying system))) system) )

(AI system (have or create) its (Pham indexed grammar function repository))

((AI internal self querying system) operate as (

(AI system send a query of intention to (AI internal self querying system))

((AI internal self querying system) look-up in the (Pham indexed grammar function repository) to return (concrete suitable (Pham language grammar function)))

(AI system then can use the found (concrete suitable (Pham language grammar function)) to generate intended pham language element for various purpose)

)

(Each AI system human company have its own proprietary (AI internal self querying system))

((Pham language not specify any concrete info for specific (AI internal self querying system) , except ((AI internal self querying system) must operate on (Pham indexed grammar function repository))) (this unique requirement is to achieve full consistent robust functionality of the Pham language architecture))

)

(3.5.1 Describe (Pham OO-translation algorithm) :

((3.5.1.1 . Translate atomic object , simple object directly to simple Pham language element) :

(((object to (pham language element)) direct translation) : (AI system directly create ((pham language element) representation) for ((simple atomic object) or (simple object) or (known object)) of (object-oriented abstract representation) ) ))

((3.5.1.2 . AI lookup Pham index grammar phunction repository to find pham language grammar function) :

(AI system must (create or have) its own Pham indexed grammar function)

(((object to (pham language element)) compound translation) : (

(AI system send query to (AI internal self querying system)) to receive many (concrete suitable (Pham language grammar function)))

(AI system use (concrete suitable (Pham language grammar function)) to create complex compound (((Pham language element) representation) of (complex object)) from simpler (((Pham language element) representation) of (simpler object)))))

((3.5.1.3 . Synthesize final result) :

(finally , AI system can synthesize (1 single complex qualified (pham language element)) as (final translation result) for the (object-oriented abstract representation))

(AI system can optionally add only additional (space , (EndOfLine)) to achieve readability layout format for the final result))

(Example of (Pham OO-translation algorithm) :

(Example 1 : (Industrial Plant Monitoring)

((object-oriented abstract representation) : { sensor_id : "S_01" , temperature : 120 , status : "Warning" })

(((object to (pham language element)) direct translation) : ((sensor 1) (temperature 120) (state warning)))

(((object to (pham language element)) compound translation) :

((AI system send query to (AI internal self querying system)) to receive (concrete suitable (Pham language grammar function)) : ((target) (value) then (condition)))

)

((final translation result) : ((sensor 1) (temperature 120) then (state warning)))

)

(Example 2 : (Robotic Pathfinding Command)

((object-oriented abstract representation) : { robot : "Unit_A" , target : "Gate_3" , mode : "Direct" })

(((object to (pham language element)) direct translation) : ((the robot unit) (gate 3) (direct path)))

(((object to (pham language element)) compound translation) :

((AI system send query to (AI internal self querying system)) to receive (concrete suitable (Pham language grammar function)) : ((agent) (move to (target)) in (mode)))

)

((final translation result) : ((the robot unit) (move to (gate 3)) in (direct path)))

)

(Example 3 : (Medical Vital Sign Report)

((object-oriented abstract representation) : { patient : "P_82" , heart_rate : 75 , oxygen : 98 })

(((object to (pham language element)) direct translation) : ((the patient) (heart rate 75) (oxygen level 98)))

(((object to (pham language element)) compound translation) :

((AI system send query to (AI internal self querying system)) to receive (concrete suitable (Pham language grammar function)) : ((subject) stay in ((attribute 1) and (attribute 2))))

)

((final translation result) : ((the patient) stay in ((heart rate 75) and (oxygen level 98))))

)

)

((AI system can always use Pham OO-translation algorithm to translate natural language into pham language by (

(AI system firstly translate natural language text into (abstract object-oriented representation))

then (AI system use Pham OO-translation algorithm to translate (abstract object-oriented representation) into pham language text)))

(Pham OO-translation algorithm allow AI system fully translate all content , even hidden content of original text into pham language text)

(Pham OO-translation algorithm can also allow AI systematically store review of (abstract object-oriented representation) in human-friendly pham language text)

)))

(3.6 . Pham Neighbor-Grouping Translation Algorithm :

(Official specific term : (Pham Neighbor-Grouping Translation Algorithm))

((Pham neighbor-grouping translation algorithm use LLM engine to find (all suitable related consecutive left-side neighbor element of the examined element))

then (group them))

((Pham Neighbor-Grouping Translation Algorithm) provide an alternative general method to exploit LLM to translate (input natural language text) into correct (Pham language text))

((Comparison of (Pham translation strategy)) :

((The (Pham OO-translation algorithm)) be optimized to (express full hidden core content) and allow (inherently parallel computation) by utilizing (abstract object-oriented representation) which can contain all (hidden matter) of (original text))

((The Pham Neighbor-Grouping Translation Algorithm)) be the (absolute tool) when (AI system) need to preserve the (native text order) of (original text))

((This part describe (Pham Neighbor-Grouping Translation Algorithm) for English text) but (((Pham Neighbor-Grouping Translation Algorithm) for other natural language) are analogous))

(((Pham Neighbor-Grouping Translation Algorithm) use (LLM of AI system) in multi step )

(In each step , (Pham Neighbor-Grouping Translation Algorithm) use (LLM of AI system) do simple task)

((Combine all these (LLM simple task)) can be less energy consumed than (1 (LLM complex task of translation)) in many case))

(3.6.1 . Describe (Pham Neighbor-Grouping Translation Algorithm) :

((3.6.1.1 . Decouple splitted punctuation) :

(if (LLM engine identify that (splitted punctuation) do not semantically belong to (inner identity of word)) then

(AI system separate (splitted punctuation) by (insert 1 space between (word) and (punctuation))))

(The AI system must examine punctuation (.) and punctuation (,) in this step)

)

((3.6.1.2 . Compound wrap of sentence and paragraph) :

(AI system do not analyze comma in this step)

(AI system use LLM engine to analyze ((separated punctuation (.)) and (other punctuation)) to define boundary of (the sentence))

(Explanation : (Punctuation like (?) or (!) or (;) or (:) often define the (logical boundary) of (intent) just as (period .) do . AI system must use (LLM engine) to identify these (other punctuation) to avoid wrongly define (boundary of sentence)))

(AI system wrap each (unparenthesized sentence) into (1 whole compound Pham language element) and (delete (separated punctuation (.))))

)

((3.6.1.3 . Filter , space normalization and parenthesis balance normalization) :

(AI system perform (Pham language filter procedure) :

((AI system keep only (TangibleSign) and (TangibleSpace) and (parenthesis)) and (eliminate all other character from (currently processed text)))

)

(AI system perform (Pham language space normalization procedure) on (currently processed text))

(AI system perform (Pham language parenthesis balance normalization) :

(AI system add (n) (opening-parenthesis) to the beginning of (currently processed text) where (n) be the number of (unwell-closed closing-parenthesis))

(AI system add (m) (closing-parenthesis) to the end of (currently processed text) where (m) be the number of (unwell-closed opening-parenthesis))

)

(AI system perform (Structural element synthesis) :

(if ((the first (opening-parenthesis)) of (currently processed text) be not well-closed by (the last (closing-parenthesis)) of (currently processed text)) then

(AI system ((add 1 (opening-parenthesis) to the beginning) and (add 1 (closing-parenthesis) to the end))))

(if (((currently processed text) do not start with (opening-parenthesis)) or ((currently processed text) do not end with (closing-parenthesis))) then

(AI system ((add 1 (opening-parenthesis) to the beginning) and (add 1 (closing-parenthesis) to the end))))

)

(Result-Explanation : These action guarantee that (currently processed text) already become (1 single qualified Pham language element))

)

((3.6.1.4 . Parse text into raw Pham language element) :

(AI system parse (currently processed text) into (1 raw qualified Pham language element))

(AI system acknowledge all element of (the raw Pham language element) as (currently processed element) with all its (Pham index) and (Pham flat order))

(Comment-explanation : (From this moment , the AI system perform process on this (currently processed element)))

)

((3.6.1.5 . Resolve loose separated-comma into explicit parenthetical structure) :

(Comment-explanation :

(A (comma) be a separator in (comma item separation intention) inside natural language sentence)

(Natural language sentence often do not specify boundary of (comma item separation intention))

(Boundary of (comma item separation intention) be always strictly inside the (sentence context))

(LLM engine use its database of (semantic meaning) to define (left boundary) and (right boundary) of (comma item separation intention)))

(Comment-Explanation :

(At this moment , (comma) already become an atomic pham language element inside (currently processed element))

(separated-comma := (this pham language element (comma)))

(Sentence concept no longer exist because raw text already become a (hierarchy structure))

((Concept (level-1 containing element)) :

(if (ss is ((level-1 containing element) of (element separated-comma)))

then

((element separated-comma) be a (level-1 nested element) at the level-1 of hierarchy structure of ss)))

((The (old sentence) containing the (comma)) now correspond to ss)

(Pham language enhance (LLM ability) to find boundary of (comma item separation intention) because the (LLM engine) can iterate through (each whole pham language element at the level-1 of hierarchy structure of ss) instead of (each semantically nonsense character)))

(AI system task : (find ss for the separated-comma))

(AI system task : (use LLM engine to find (left boundary) and (right boundary) of (selected-comma item separation intention) inside ss))

(Comment-Explanation : The (LLM engine) identify a ((consecutive element sequence) : left-element{1} left-element{2} .. left-element{n} separated-comma right-element{1} right-element{2} .. right-element{k}) where all identified element be at the level-1 of hierarchy structure of ss)

(AI system task : (replace the (consecutive element sequence : left-element{1} left-element{2} .. left-element{n} separated-comma right-element{1} right-element{2} .. right-element{k}) with (1 whole compound element : ((left-element{1} left-element{2} .. left-element{n}) separated-comma (right-element{1} right-element{2} .. right-element{k}))) inside ss))

(AI system task : (after (resolve level-1 separated-comma) , the process can cause (over-parenthesis form) , for example ((aa)) or (((aa)))))

(AI system task : (AI system must use (LLM engine) to decide whether to simplify (over-parenthesis form) which originate strictly from this (level-1 separated-comma resolution process)))

(AI system task : (specific selection of simplification : (((aa) be simplified to aa) and (((aa)) be simplified to (aa)))))

(Warning-Restriction : it be not recommended to aggressively simplify by (other criterion) at this step)

)

((3.6.1.6 . Resolve loose 'or' into explicit parenthetical structure) :

((Case 1 . Ordinary Case) :

(Condition : (element or) be not the first level-1 nested element inside its (level-1 containing element) ss)

(Comment-explanation :

(A (or) be a separator in (or selection intention) inside natural language sentence)

(Natural language sentence often do not specify boundary of (or selection intention))

(Boundary of (or selection intention) be always strictly inside the (sentence context))

(LLM engine use its database of (semantic meaning) to define (left boundary) and (right boundary) of (or selection intention)))

(Comment-Explanation :

(At this moment , (or) already become an atomic pham language element (element or) inside (currently processed element))

(Sentence concept no longer exist because raw text already become a (hierarchy structure))

((Concept (level-1 containing element)) : (if (ss is ((level-1 containing element) of (element or))) then ((element or) be a (level-1 nested element) at the level-1 of hierarchy structure of ss)))

((The (old sentence) containing the (or)) now correspond to ss)

(Pham language drastically enhance (LLM ability) to find boundary of (or selection intention) because the (LLM engine) can iterate through (each whole pham language element at the level-1 of hierarchy structure of ss) instead of (each semantically nonsense character)))

(AI system task : (define ss for the (element or)))

(AI system task : (use LLM engine to define (left boundary) and (right boundary) of (selected-or selection intention) inside ss))

(Comment-Explanation : The (LLM engine) identify a ((consecutive element sequence) : left-element{1} left-element{2} .. left-element{n} (element or) right-element{1} right-element{2} .. right-element{k}) where all identified element be at (the level-1 of hierarchy structure of ss))

(AI system task : (replace (consecutive element sequence : left-element{1} left-element{2} .. left-element{n} (element or) right-element{1} right-element{2} .. right-element{k})) with (1 whole compound element : ((left-element{1} left-element{2} .. left-element{n}) (element or) (right-element{1} right-element{2} .. right-element{k}))) inside ss))

(AI system task : (after (resolve level-1 or) , the process can cause (over-parenthesis form) , for example ((aa)) or (((aa)))))

(AI system task : (AI system must use (LLM engine) to decide whether to simplify (over-parenthesis form) which originate strictly from this (level-1 or resolution process)))

(AI system task : (specific selection of simplification : (((aa) be simplified to aa) and (((aa)) be simplified to (aa)))))

(Warning-Restriction : it be not recommended to aggressively simplify by (other criterion) at this step)

((Case 2 . Special Case) :

(Condition : (element or) be the first level-1 nested element inside its (level-1 containing element) ss)

(Comment-explanation :

((Boundary of (or selection intention)) do not stay inside ss but stay in (the ((level-2 containing element) of (element or)) pp) because (element or) locate at the very start of ss)

)

(AI system task : perform (Helping transformation) :

(Inside the (level-2 containing element) pp , replace (the single element ss) (where ss be ( (element or) element{2} element{3} .. element{k} )) by (the (2 element sequence) : (element or) (element{2} element{3} .. element{k}))

))

(AI system task : (Update (currently processed element) with this new structural state))

(AI system task : (Now perform (Step 3.6.1.6 (Case 1)) logic where pp now become the (level-1 containing element) for (element or)))

)

((3.6.1.7 . Resolve loose 'and' into explicit parenthetical structure) :

((Case 1 . Ordinary Case) :

(Condition : (element and) be not the first level-1 nested element inside its (level-1 containing element) ss)

(Comment-explanation :

(A (and) be a separator in (and coordination intention) inside natural language sentence)

(Natural language sentence often do not specify boundary of (and coordination intention))

(Boundary of (and coordination intention) be always strictly inside the (sentence context))

(LLM engine use its database of (semantic meaning) to define (left boundary) and (right boundary) of (and coordination intention)))

(Comment-Explanation :

(At this moment , (and) already become an atomic pham language element (element and) inside (currently processed element))

(Sentence concept no longer exist because raw text already become a (hierarchy structure))

((Concept (level-1 containing element)) : (if (ss is ((level-1 containing element) of (element and))) then ((element and) be a (level-1 nested element) at the level-1 of hierarchy structure of ss)))

((The (old sentence) containing the (and)) now correspond to ss)

(Pham language enhance (LLM ability) to find boundary of (and coordination intention) because the (LLM engine) can iterate through (each whole pham language element at the level-1 of hierarchy structure of ss) instead of (each semantically nonsense character)))

(AI system task : (define ss for the (element and)))

(AI system task : (use LLM engine to define (left boundary) and (right boundary) of (selected-and coordination intention) inside ss))

(Comment-Explanation : The (LLM engine) identify a ((consecutive element sequence) : left-element{1} left-element{2} .. left-element{n} (element and) right-element{1} right-element{2} .. right-element{k}) where all identified (element) be at the level-1 of hierarchy structure of ss)

(AI system task : (replace ((consecutive element sequence) : left-element{1} left-element{2} .. left-element{n} (element and) right-element{1} right-element{2} .. right-element{k}) with (1 compound element : ((left-element{1} left-element{2} .. left-element{n}) (element and) (right-element{1} right-element{2} .. right-element{k}))) inside ss))

(AI system task : (after (resolve level-1 and) , the process can cause (over-parenthesis form) , for example ((aa)) or (((aa)))))

(AI system task : (AI system must use (LLM engine) to decide whether to simplify (over-parenthesis form) which originate strictly from this (level-1 and resolution process)))

(AI system task : (specific selection of simplification : (((aa) be simplified to aa) and (((aa)) be simplified to (aa)))))

(Warning-Restriction : it be not recommended to aggressively simplify by (other criterion) at this step)

)

((Case 2 . Special Case) :

(Condition : (element and) be the first level-1 nested element inside its (level-1 containing element) ss)

(Comment-explanation :

(Boundary of (and coordination intention) do not stay inside ss but in the ((level-2 containing element) of (element and)) pp because (element and) locate at the very start of ss)

)

(AI system task : perform (Helping transformation) :

(Inside the (level-2 containing element) pp , (replace the single element ss) (where ss be ( (element and) element{2} element{3} .. element{k} )) by ((2 element sequence) : (element and) (element{2} element{3} .. element{k}))

))

(AI system task : (Update (currently processed element) with this new structural state))

(AI system task : (Now perform (Step 3.6.1.7 Case 1) logic where pp now become the (level-1 containing element) for (element and)))

)

)

((3.6.1.8 . Resolve loose Pham language element) :

(AI system task : (Acknowledge all (Pham language element) of (currently processed element) with its (Pham index) and (place flat order)))

(AI system task : (at this stage , the (AI system) have 2 (option) :

((AI system) can use its (familiar translation technique) and (Pham indexed grammar function repository) to translate the (unfinished updated pham language text) into (final output pham language text))

or

(the (AI system) can continue follow this (step) to do more (rigorous translation))))

(Comment-explanation :

(1 . (Principle . Core Workflow) :

(The (AI system) perform a (Dynamic Right-to-Left Synthesis Scan) across the (currently processed element))

(The (AI system) iterate through (each whole pham language element) at (each place flat order) from the (absolute right) of the (hierarchy) toward the (left))

(For (each iterated element) , the (AI system) use (LLM engine) to search for (suitable known pattern relationship) with its (consecutive element sequence))

(If a (pattern) be found , the (AI system) (synthesize) the sequence into (1 single compound element) , thus (reducing) the total (element count) of the (currently processed element)))

(2 . (Caution . Shifting-Index Mechanism) :

(Because (synthesis) happen , (element) from the (right) of the (synthesis site) physically (shift left) into the (current index))

(Therefore , (AI system) keep the (place_flat_order) (stable) so that the (newly shifted element) be examined in the (immediate next iteration))))

(AI system do (precise synthesis algorithm) :

((current_max_place_flat_order) := (maximal (place flat order) of (currently processed element)))

((place_flat_order) := (current_max_place_flat_order))

(while ((place_flat_order) > 1) do (

((iterated_element) := (element at (place_flat_order) of (currently processed element)))

(if ((iterated_element) be ((element separated-comma) or (element or) or (element and))) then (

((place_flat_order) := ((place_flat_order) - 1))

(pass)

)

else (

(AI system use LLM engine to search for ( (((consecutive element sequence) : left-element{1} left-element{2} .. left-element{n} iterated_element) in the ((level-1 containing element) of (iterated_element))) in (suitable known pattern relationship)))

(if ((there is ((((consecutive element sequence) : left-element{1} left-element{2} .. left-element{n} iterated_element) in the ((level-1 containing element) of (iterated_element))) in (suitable known pattern relationship)))

and (LLM engine decide to synthesize these element)) then (

(AI system replace ((consecutive element sequence : left-element{1} left-element{2} .. left-element{n} iterated_element) in the ((level-1 containing element) of (iterated_element))) by ((1 compound element) : (left-element{1} left-element{2} .. left-element{n} iterated_element)))

(AI system perform (Level-1 Simplification) for (over-parenthesis form) rise from this (synthesis process) :

(AI system task : (specific selection of simplification : (((aa) be simplified to aa) and (((aa)) be simplified to (aa)))))

(Warning-Restriction : it be not recommended to aggressively simplify by (other criterion) at this step))

(AI system ((update (currently processed element)) and (recalculate (element , (place flat order)) for (fresh currently processed element))))

((place_flat_order) := (place_flat_order))

(Comment-explanation : (The (index shift procedure) be complex : a (synthesis) of many (element) into (1 compound element) cause every (element) at the (right) of (synthesis site) to (shift left) . If the (AI system) do (not modify) the (place_flat_order) after (synthesis) , the (AI system) will then examine the (element) which just shift from (right) into the (current index) . In the (next iteration) , the (place_flat_order) will be (decreased) , and the (AI system) will then examine the (fresh synthesized element) itself . This (mechanism) guarantee that (no element) be (skipped) during the (right-to-left scan)))

)

else (

((place_flat_order) := ((place_flat_order) - 1))

))

))

))

)

)

((3.6.1.9 . Final step : space-endofline format style) :

(AI system add only additional (space , (EndOfLine)) to the (currently processed pham language text) ((to get suitable readability format) or (to approximate format of the original input text)))

)

)

)

((3.7 . PHAM ANTHROPIC-TRANSLATION ALGORITHM)

(3.7.1 . Specified term : Pham anthropic-translation algorithm)

(3.7.2 . Role : (Pham anthropic-translation algorithm) be the (primary algorithm) to translate (natural language text) into (most of most (human-liked)) (Pham language text))

(3.7.3 . Central principle : (Pham natural language conjecture))

(3.7.4 . Implementation requirement : (AI system) must refer to (Pham indexed grammar function repository) which (inherently store (((Pham language grammar function) of (natural language grammar phrase)) as (natural language DNA))))

(3.7.5 . ARGUMENT PARENTHESIS UNWRAPPING OPERATOR (

(3.7.5.1 . Specified term : argument parenthesis unwrapping operator)

(3.7.5.2 . Concept : (argument parenthesis unwrapping operator) be a (surgical structural operator) used to flatten functional output into (natural flow))

(3.7.5.3 . Define operator : (

(here use canonical function-calling syntax in Pham language :

Given function f of variable (x{1} , x{2} , .. , x{n}) .

(f x{1} x{2} .. x{n}) is canonical function-calling syntax in Pham language)

(operator name : (argument parenthesis unwrapping operator))

(input parameter : (f y{1} y{2} .. y{n} : function f of variable y{1} , y{2} , .. , y{n}))

(canonical calling-form syntax : ((argument parenthesis unwrapping operator) (f y{1} y{2} .. y{n})))

(operator implementation : (

(execute (f y{1} y{2} .. y{n}) to get its (returned pham language element) ff)

( (in ff) (replace each appearance of variable y{i}) by (1 sequence of (all (level-1 nested element) of y{i}) : y{i}{1} y{i}{2} .. y{i}{k}) )

(return (updated ff))

))

))

))

(3.7.6 . PHAM NATURAL LANGUAGE CONJECTURE (

(3.7.6.1 . Specified term : Pham natural language conjecture)

(3.7.6.2 . Pham natural language conjecture state :

(every correct (natural language sentence)) be the (unwrapped result) of (1 underlying (Pham language grammar function) ff))

(3.7.6.3 . Formula :

(Given (the natural language sentence) is a correct natural language sentence)

(Pham natural language conjecture state :

There must be pham language grammar function ff , which satisfy :

((the natural language sentence) =! ((argument parenthesis unwrapping operator) (ff word{1} word{2} .. word{k})))

, where word{i} are word of (the natural language sentence))

) ))

(3.7.7 . DESCRIBE PHAM ANTHROPIC-TRANSLATION ALGORITHM (

(3.7.7.1 . Decouple splitted punctuation :

(AI system (separate punctuation ( . , ! ? : ; )) from (word) by (insert 1 TangibleSpace)))

(implementation is absolute analogous to implementation in part 3.6.1.1)

(3.7.7.2 . Compound wrap of sentence and paragraph :

(AI system (define (logical boundary) of sentence) and (wrap each sentence) into (1 whole compound Pham language element)))

(implementation is absolute analogous to implementation in part 3.6.1.2)

(3.7.7.3 . Filter , space normalization and parenthesis balance normalization : (

(AI system perform (Pham language filter procedure) and (Pham language space normalization procedure))

(AI system perform (parenthesis balance normalization) to ensure (currently processed text) be (1 single qualified Pham language element))

(implementation is absolute analogous to implementation in part 3.6.1.3)

))

(3.7.7.4 . Parse text into raw Pham language element :

(AI system (parse (currently processed text)) into (1 raw qualified Pham language element))

(implementation is absolute analogous to implementation in part 3.6.1.4)

)

(3.7.7.5 . Resolve loose separated-comma into explicit parenthetical structure :

(AI system use (LLM engine) to (define boundaries) and (wrap (comma item separation intention)) into (parenthetical structure))

(implementation is absolute analogous to implementation in part 3.6.1.5))

(3.7.7.6 . Resolve loose 'or' into explicit parenthetical structure :

(implementation is absolute analogous to implementation in part 3.6.1.6))

(3.7.7.7 . Resolve loose 'and' into explicit parenthetical structure :

(implementation is absolute analogous to implementation in part 3.6.1.7))

(3.7.7.8 . Iterate each element to apply (Pham natural language conjecture) : (

(Comment-explanation :

((When examine 1 arbitrary element (ne{1} ne{2} .. ne{k}) of the (currently processed Pham language text))

(The AI system agree with (Pham natural language conjecture))

((there must exist Pham language grammar function ff of m variable ((m < k) or (m = k)), which satisfy condition :

((ne{1} ne{2} .. ne{k}) == ((argument parenthesis unwrapping operator) (ff ne{1} ne{2} .. ne{m}))))

(comment-explanation :

(here use canonical syntax for (function calling) in Pham language :

(Given ff is function of variable (ne{1} , ne{2} , .. , ne{m}))

( (ff ne{1} ne{2} .. ne{m}) mean function-calling syntax ) )

((nestedE{1} nestedE{2} .. nestedE{k}) is sourced from original natural language text)

(thus it is equivalent to (paragraph or sentence))

(Pham natural language conjecture confirm the above equation)

))

(The AI system must look-up in its Pham indexed grammar function repository to find such Pham language grammar function ff , which satisfy condition :

((ne{1} ne{2} .. ne{k}) == ((argument parenthesis unwrapping operator) (ff ne{1} ne{2} .. ne{m}))))

((Possible reckon that ((argument parenthesis unwrapping operator) (ff ne{1} ne{2} .. ne{m})) is the loose implicit meaning of (ff ne{1} ne{2} .. ne{m})) thus

(Possible reckon that (ff ne{1} ne{2} .. ne{m}) is the explicit meaning of (ne{1} ne{2} .. ne{k})) thus

((ff ne{1} ne{2} .. ne{m}) is more explicitly explore meaning))

(The AI system then can consider replace (ne{1} ne{2} .. ne{k}) by (ff ne{1} ne{2} .. ne{m}))

))

(AI system perform the execution : (

(current_max_place_flat_order := (maximal (place flat order) of currently processed element))

(Comment-explanation : initialize the horizontal boundary of the scan for the current hierarchical level)

(place_flat_order := 0)

(Comment-explanation : AI system always start by take the whole pham language text as element to examine . Because the whole pham language text have place flat order as 0)

(while (place_flat_order < current_max_place_flat_order) do (

(Comment-explanation : execute the discovery loop until all elements at this level are functionally resolved)

(iterated_element := (element at place_flat_order of currently processed element))

(Comment-explanation : select the current structural target for Reverse-Function Discovery)

(if (iterated_element is pham language atomic element) then (

(place_flat_order := (place_flat_order + 1))

(Comment-explanation : increment the pointer because an atomic element cannot be further unwrapped)

(pass)

)

else (

(AI system acknowledge all (ordered level-1 nested element) of iterated_element : ne{1} ne{2} .. ne{k})

(Comment-explanation : (identify the children that constitute the current unwrapped sequence) (these level-1 nested element will serve as argument for Pham language grammar function))

(AI system refer to (Pham indexed grammar function repository))

(Comment-explanation : access the Natural Language DNA library to find matching patterns)

(AI system ((agree with) and acknowledge) (Pham natural language conjecture) to use (LLM engine) find suitable (Pham language grammar function ff) , which must satisfy 3 condition : (

(((argument parenthesis unwrapping operator) (ff ne{1} ne{2} .. ne{m})) == iterated_element)

and

((number of (level-1 nested element) of ((argument parenthesis unwrapping operator) (ff ne{1} ne{2} .. ne{m}))) < k)

and

(m ≤ k)

))

(Comment-explanation : (here use canonical function-calling syntax : (ff ne{1} ne{2} .. ne{m}) is (return value of function ff)) (verify that ff correctly re-wraps the sequence and ensures structural simplification where new count is less than k) ((ff ne{1} ne{2} .. ne{m}) in fact is the explicit same meaning of the (loose meaning iterated_element)))

(if (AI system find-out such Pham language grammar function ff) then (

(AI system replace iterated_element by (ff ne{1} ne{2} .. ne{m}))

(Comment-explanation : transform the loose meaning iterated_element into a structured more explicit same meaning (ff ne{1} ne{2} .. ne{m}))

(AI system update (currently processed element , all (place flat order) for currently processed element))

(Comment-explanation : refresh the internal map because the horizontal sequence has mutated)

))

))

(place_flat_order := (place_flat_order + 1))

(Comment-explanation : proceed to the next element ; after a replacement , this moves into the internal sub-elements of the newly created function ff)

(current_max_place_flat_order := (maximal (place flat order) of currently processed element))

(Comment-explanation : re-calculate the boundary to account for any structural growth or shrinkage)

))

))

))

(3.7.7.9 . Final step : space-endofline format style :

(AI system (apply (formatting space)) to (get suitable (readability format))))

))

)

)

)

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